



BRIAGE User Manual

- Again with an unsettling colour scheme! -

Documentation - EN - v2.5.786

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for Blender 2.7x and Train Simulator 20xx

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1 Foreword

1.1 Thanks

Thanks a lot to:

- Dom107
- JADsHome
- Karma99(Victory Works)
- Mike Adams
- Hugo_B
- Jáchym Hurtík

for their remarks, tests and help around the exporter!

All of this wouldn't have been possible without Dom107's exporter anyway.

1.2 Personal thoughts about the exporter

This new exporter is being worked on since November 2015. I am very proud of the result and having been part of the great adventure of this exporter which contributes to the successful DTG's Train Simulator 20xx series (I'm not affiliated to Dovetail Games in any ways).

I still don't know why anybody made this before, but I'm happy it's here for you now.

I began working on this because using the old exporter was too extensive in time and resources compared to the results. I also wanted it to be more ergonomic for newcomers to 3D and complex TS asset authoring.

I learnt Python on-the-job for this. After 2 months I got something more useful through blender's custom properties, this was the V1.5 . More internal versions came by while I was getting better and hence quicker with Python. More and more features got implemented too until it may finally be named a "V2.0". Some corrections and hundreds of revisions later, there was finally something with most of the features I wanted and a stable core. That's where few of you began hearing of it on *UKTS* or *RailSim-fr.com* some talented beta-testers applied and two months later quite everything was finished thanks to their remarks!

Thanks to them you can now use a mostly bug-free exporter! (according to test's extrapolations....)

Why so late? I wanted it be the best I can give to the community. I typed this thing alone in my bedroom all day and everyday for months, and used it as an excuse to cast aside as much people as I could to stay alone in my little PC world (which was very effective). And finally, this was really hard sometimes. You don't believe me? -Ask my father or try yourself to develop something based on someone else work in a language you don't understand!-

Fun fact, I always thought I was alone in this development ship, but in facts, there where lots of people supporting me. Sparse PMs or half-weekly phone calls: everything and everyone weighs in the scale! They don't belong in the "special thanks" section but they will recognize themselves while reading this (if they ever read it... hahaha). This was for me a rewarding programming and typing experience as much as an enriching human experience in the end.

My lasts words about this are that I gladly wish this software will be as useful to you as it was to me.

1.3 Where does that name comes from?!

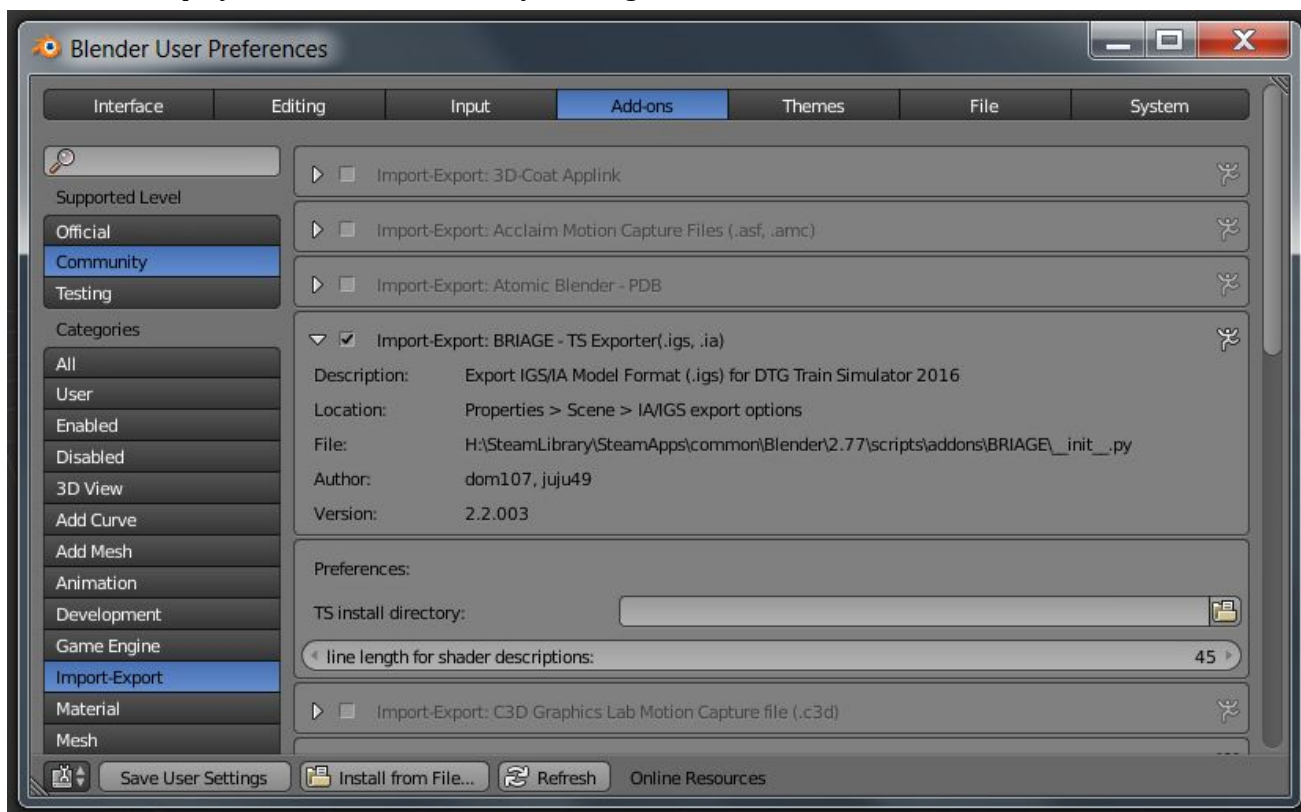
BRIAGE is an acronym for **BlendeR Intermediate Animation and Geometry Exporter** which is the full name of this software.

It's also an old French word which designates the last step of work where a dough or pastry is homogenized and packed using a wooden bar or a mechanical device. I found this to be an appropriate enough fact for that name to be kept compared with others I came up with so far.

2 Installation

2.1 Install files

- Select your installation method:
 - With Windows Explorer:
 - Install BRIAGE Add-On by simply copying “BRIAGE” folder in Blender “scripts\addons” sub-directory (for example: `D:\Program Files\Blender Foundation\Blender\2.63\scripts\addons`).
 - Start Blender to open the default Blender file.
 - Open **Files > User Preferences...**
 - go to **Add-ons** tab.
 - Look for “BRIAGE – TS Exporter(.igs, .ia)”.
It is advisable to filter add-ons for this purpose. Select:
 - Supported Level: “Community”
 - Categories: “Import-Export”
 - With Blender Filesystem explorer
 - Start Blender to open the default Blender file.
 - Open **Files > User Preferences...**
 - go to **Add-ons** tab.
 - Click **Install from file** button. A file browser opens.
 - Reach “BRIAGE.zip” file and confirm.
- The installed Add-On must be enabled before they can be used. Simply place a check mark on the **Enable Add-On** box. The new export functionalities are now integrated into Blender.
- Display add-on informations by clicking the white arrow near the checkbox.



4. Specify TS main “\Railworks” folder path (where TS executable file is located). This is used to reach shader information.
If not set, it will be regenerated on each load from system and steam informations. However this last technique might fail in some cases and an error will be displayed in material IGS settings shader description box.
5. Specify a line length in characters per line for shader description box text if necessary. This option allows you to make sure you can see shader descriptions.
6. Click on “Save User Settings”.

2.2 Uninstall files

1. Start Blender to open the default Blender file.
2. Open Files > User Preferences...
3. go to Add-ons tab.
4. Look for “BRIAGE – TS Exporter(.igs, .ia)”.
It is advisable to filter add-ons for this purpose. Select:
 - Supported Level: “Community”
 - Categories: “Import-Export”
5. Simply uncheck its “Enable Add-On” box . The new export functionalities are now withdrawn from Blender.
6. Select the same as your installation method:
 - With Windows Explorer:
 1. Delete “BRIAGE” folder in Blender “scripts\addons” sub-directory (for example: D:\Program Files\Blender Foundation\Blender\2.63\scripts\addons\BRIAGE).
 - With Blender Filesystem explorer
 1. Display add-on informations by clicking the white arrow near the checkbox.
 2. Click “remove” button

3 Presentation

3.1 Contents

The exporter is made of 2 main parts:

- IGSE module: IGS format export (model structure)
- IAE module: IA format export (animation)

You, the user, influence its behaviours through Blender's interfaces mainly located in the properties area:

- **Material tab panels (see 6)**
- **Scene tab panels (see 4)**
- Texture tab panels (see 7)
- User Preferences Add-ons tab panel (see 2)
- Info tab on long exports to display a progress bar

Some example files are provided too in the package.

3.2 What is processed from Blender?

- ✓ **Objects**
 - +positions
 - +rotations
 - +main object colour
- ✓ **Meshes**
 - +materials
 - +vertices
 - +faces
 - +custom normals
 - +active Vertex colours (unlike in Blender only one colour is saved per vertex), see 8.7.3
 - +Vertex groups weights (if linked to bones through armature modifier), see 8.3
- ✓ **Bones**
 - +positions
 - +rotations
 - +deform bindings up to 4 bone influences per vertex
- ✓ **Meta** objects, **Curves** (if geometry is found), **Texts** and **NURBS** can be exported too as regular objects if they have 1 material assigned. They are auto-converted during export to regular objects and your scene stays untouched.
- ✓ **All dupli** counterparts of the preceding statements

3.3 Usual workflow

The principle to create an exportable model is as follows:

Name objects

An object must follow the TS20xx strict naming rules for LODs (see in 10.4).

If an object doesn't follow these rules, some processing is made to try to set the name as required by the game (see in 10.4).

Texture meshes

A material usually needs several texture slots, their content depends on the shader used (see in 7.1, 10.1, 10.2).

Adjust some pivots

Outside specific TS20xx or animations requirements, it is advisable to centre the pivot of each object on its geometry.

Among the specific requirements of TS20xx, the pivot of the main body of a car or locomotive must be centred with $z = 0$ at the rail contact (See example in 4.7). The bogie pivot must coincide with its rotation axis and the bogies naming convention must be followed with bo01 as the front bogie and bo02 as the back bogie, in the case of a rolling stock with 2 bogies.

Setup material attributes in the Blender material tab

In Blender, you can select shader names in a dropping menu, use custom ones or even use the old still supported way through a material's first texture slot name. These two last fields support entering short names or full shader names (See 7.1, 10.1).

Note that a specific processing is performed for `TrainGlassWeatherEffects.fx` to make it more convenient to use (see 8.7). Also some automatic adjustments are done depending on the shader used. Material attributes are initialized in Blender (see 4, 6).

Create animations (See in 5, 8.5)

Setup the behaviour of the exporter (see in 4, 6, 7)

The exporter takes the modifiers into account without applying them to the model. Objects can also have some scaling and rotation (see some restrictions in 8.4 : it could be necessary to use CTRL+A to apply these transformations). Ngons can also be used.

As a default behaviour, several Blender objects are merged into one object when exported, if they all have the same LOD value and distance. This can be cancelled or altered, take a look here 4.2 and here 4.7 for more details.

Main object field (see in 4.4) sets the main object of the model, for example the body of a locomotive.

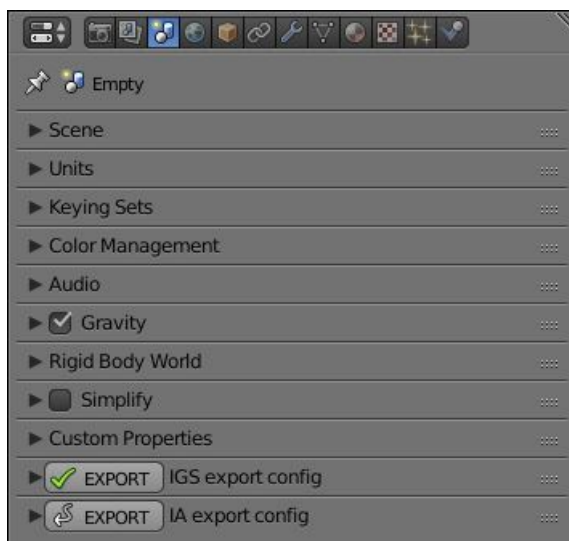
There are 2 modes of export for IGS or IA: the entire model (whether objects are selected or not in Blender views) or selected objects (see 4.2). Main object is set to active object by default.

Annexe 10.6 provides some information about using dynamic representation of character strings (automatic numbering).

Check log or Blender's system console for errors, see 8.4.2 - Reading logs

4 Configuration for IGS Export (IGSE)

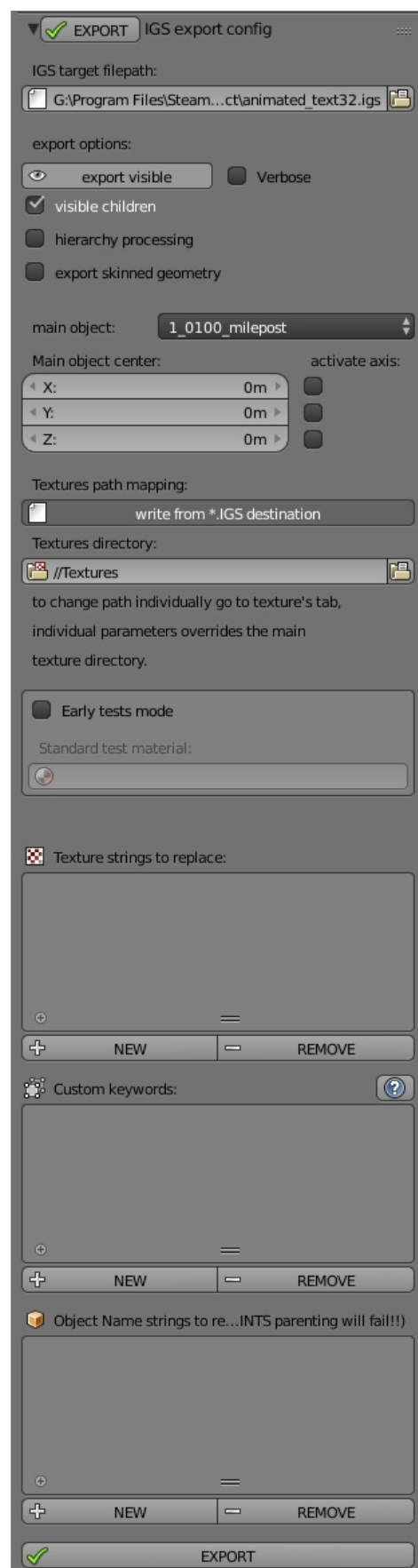
The following paragraphs explain how to configure IGSE (IGS Exporter) main options through the Properties panels:



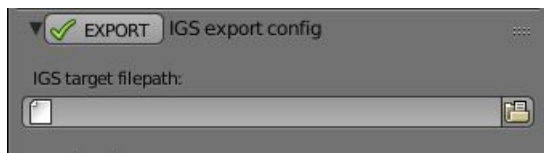
These properties are scene-dependent. Different Blender scene will not share configurations implicitly. This allows the exporter to be more flexible and *.blend files might contain different objects without parameters interfering with each other.

You can store a whole train-set in a blend, sharing materials between cars for easy editing, share bogies and doors while keeping each car in its own scene not overlapping with other train parts and each car has its own IGS Export config.

It is **highly discouraged to use and export object which don't have their scales applied**. Especially if it is a negative scale or a LOD number higher than 1. It is indeed more prone to provoke unexpected results. If used in conjunction with center main object parameters!



4.1 Main settings



The top `Export` button allows exporting to IGS format.

IGS target filepath is the last export path used by IGSE. You can enter/modify it here later after clicking `Export`. It is saved in the Blend file for later use.

4.2 Export mode and other important behaviours



The first top left box sets the `Export` mode to export selected or export visible. They are self-explanatory.

“Export selected”: only selected objects are exported with all of their children (selected or not, visible or not).

“Export visible”: only visible objects (selected or not) are exported with all of their children (visible or not).

Visible children modifies children exportation rules. When this setting is enabled, only visible children are exported whatever `Export` mode is selected.

In the *Outliner*, the eye icon in front each object is not greyed out for a visible object.

Hierarchy processing allows the exporter to merge objects together following their LOD's distance and number as well as their matching keywords. Annex 10.3 shows the impact of options `Hierarchy` and `Custom Keywords` combined on IGS and IA export. Further details at 4.7 - Keywords merging system.

For small models, it is advisable to deactivate this option.

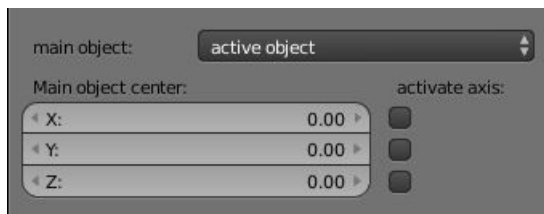
Export skinned geometry enables the export of bones and corresponding `Vertex groups` weights. See 8.3 - Using skinning and bones for informations.

4.3 Other export toggles



`Verbose` toggle makes the exporter print a full log *.igs.log file alongside your *.igs file instead of writing a short log to Blender's system console. Beware, this option increases export time by a factor of 5. Use it only when something looks wrong and you want to take a look at exported data.

4.4 Main Object settings



Main Object **dynamic enumerator** tells to select the pointed object as the one which will be used to merge other objects at 'scene level' hierarchy of the Blender model.

If “active object” is specified, the first mesh object found while scanning list of exportable Blender objects is chosen. Active object is the **yellow highlighted** object among a selection.

IGSE log indicates used main object:

```
BRIAGE:: 'PROPERTY': MainObject to find = 1_1000_BiDuLe
BRIAGE:: 'PROPERTY': MainObject found = 1_1000_BiDuLe
```

Main object center is a set of 3D coordinates with matching checkboxes. This is used to centre main object origin at these coordinates (and move children accordingly) if the corresponding checkbox is enabled. This avoids moving objects in Blender, for example, to centre a locomotive along the X axis to ensure that it is centred on the rails.

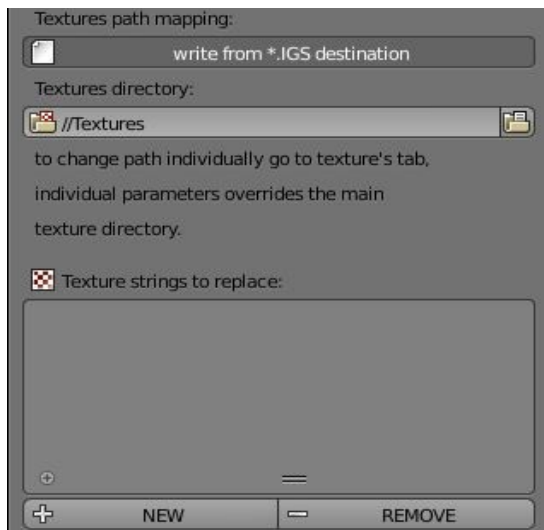
Centring along an axis gives the same result as moving the entire model in Blender so that the pivot of the Main Object object is at coordinate 0 of this axis.

*As such a move impacts animations, the IGS file **settings of “Main object center” and “Main Object” are also read when exporting IA files.***

IGSE log indicates that the model was automatically centred:

```
BRIAGE:: 'INFO': CenterMainObject: exported model moved by X=0.122961 Y=-0.211694 Z=-0.205195 (Blender axis)
```

4.5 Texture manipulation tools



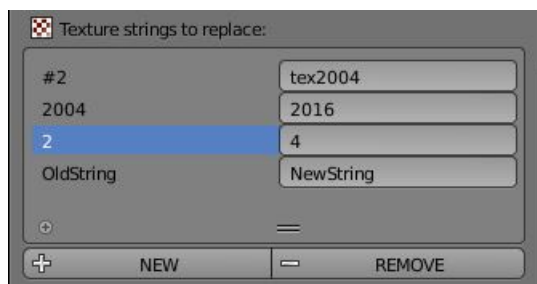
Texture path mapping mode sets how the exporter binds IGS file to texture files.

- “write from *.IGS destination” mode: wherever you export your IGS, its texture components will point to the files used/specified in Blender.

This will be referred later as “**New texture mode**”

- “write from *.blend location” mode: *the same relative filepath from blender will be applied to IGS blindly.* For this setting to work properly, work in your source folder sub-directories where you will export your IGS to or make sure your working directories mimic exactly end directories hierarchy. This will be referred later as “**Legacy texture mode**”

Textures directory field allows you to set a directory where all your model's textures will be stored directory can be child, sibling, uncle or wherever you want. By default every texture filepath is rerouted to this folder, you can fine-tune this behaviour with individual texture settings, see 7.3.



Example of use above

toggles “**texture name replacement**” mode. Each filename listed as such will be replaced by another name. The file suffix (dds, ace, bmp, etc...) must not be specified.

Example above :

Texture file “2” is changed to “tex2004”,
then “2004” is replaced by “2016”,
then “2” are translated to “4”,
finally it is exported as “tex4016”.

Settings application order:

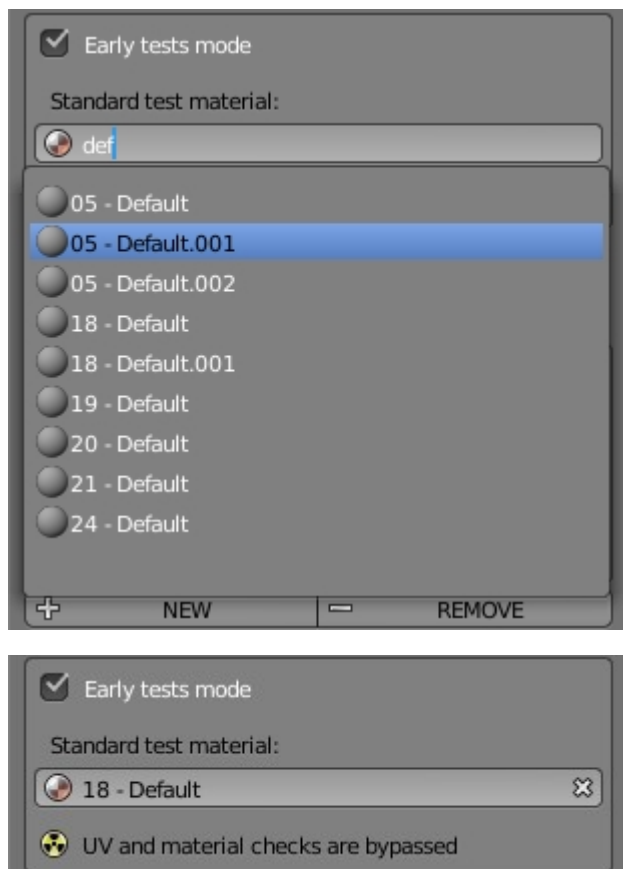
Texture strings to replace (“*replacement mode*” then “*find all mode*”) => Cancel 'Textures directory' setting **ELSE**
Export texture path **ELSE** Texture directory => Texture path mapping

Texture strings to replace is a table where you enter former (double-click to edit) and new strings to be found and replaced by in texture names.

It allows replacing each listed string by another string, for all the texture files, after “*texture name replacement mode*” has been processed. This is “*find all*” mode.

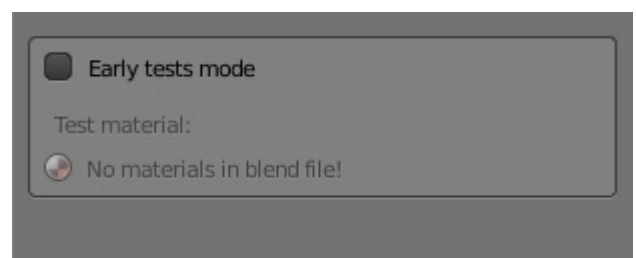
Adding “#” at the beginning of **former string field** toggles “**texture name replacement**” mode. Each filename listed as such will be replaced by another name. The file suffix (dds, ace, bmp, etc...) must not be specified.

4.6 Early tests mode



This check box enables BRIAGE to circumvent errors which may arise from exporting your models. Could it be missing or faulty materials or missing UVs, it will try to export everything it can when the checkbox is ticked.

It will nonetheless need a game-ready material in "Standard test material:" field to work. You can type in this field to narrow search results while scrolling the list.



This happens, as the text states, when there is no material in blend file

4.7 Keywords merging system

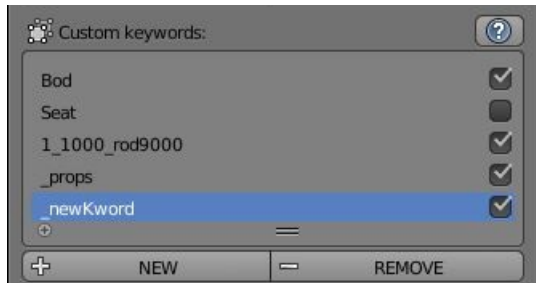
As a default behaviour, the Blender model hierarchy is kept only when objects include TS20xx keywords or when a LOD change is found. As much as possible, several Blender objects are merged into one object when exported, if they all have the same LOD value and distance. This is a recommended setting to diminish performance issues linked with a high number of direct3D "draw calls" in TS.

To merge objects, the exporter looks for the following built-in keywords. ('\$' represents a digit) :

```
'_locomotive'      '_fuel_level_'      '_door$$'
'_tender'          '_freight'          '_step$$'
'_coach'           '_bulk'             '_wh$$'
'_vehicle'         '_lights_fwdhead'   '_bo$$'
'_wagon'           '_lights_revhead'   '_panto$$'
'_carriage'        '_lights_fwdtail'   '_wiper$$'
'_coal'            '_lights_revtail'   '_primarydigits_$'
```



merging by user-defined keywords when “Hierarchy processing” is activated.



Example of use above

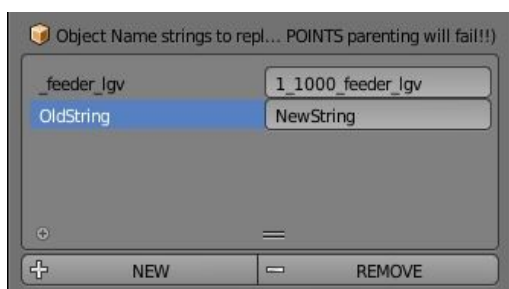
the name, those with string `_props` in the name and those with those with string `_newKword` in the name (this is default field's value).

Note that the checkbox in front of `Seat` is unticked, therefore this setting will not be used on next export until enabled again.

If “Hierarchy processing” is enabled and you do not want an animated object to be merged with other objects, unless they move with the same motion, just add its full name to the list. Thus, `1_1000_rod9000` present in Custom Keywords table will be a single object: It will not be merged with any other object.

An object named with a TS20XX keyword as listed in above blue box, (for example `1_0100_bo01`) doesn't need to be in Custom Keywords table.

4.8 Object rename

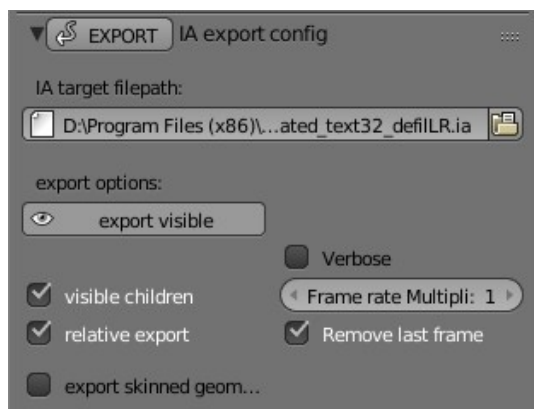


Object Name strings to replace table replaces part of an object name by right field value if left field value is found inside object name.

This feature was elaborated to be able to make complete sets of objects inside a blend file which shares alike objects which can't be named the same in blender (Ex: for dynamic gantries wire nodes: allows easier use of Asset Editor later in development process). You can use this to truncate the ".002" produced by Blender on the fly. This assumes theses alike objects do not share the same IGS file where homonymy could cause various or serious problems.

Here, every object including “`_feeder_lgv`” will have this part replaced by “`1_1000_feeder_lgv`”. This is a very bad idea: it will indeed corrupt your igs names mostly...

5 Configuration for IA Export (IAE)



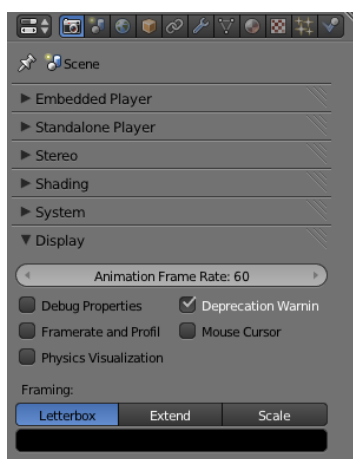
IA export config is located below IGS export config in Scene tab.

In addition to the possible use of Main object center from IGSE configuration (see 4.4), there is an optional configuration panel with two exclusive parameters: Frame rate Multiplier and Remove last frame. Other settings are stolen from IGSE; refer to it for further details 4.1, 4.2, 4.3.

See more here: 8.5 - About animation

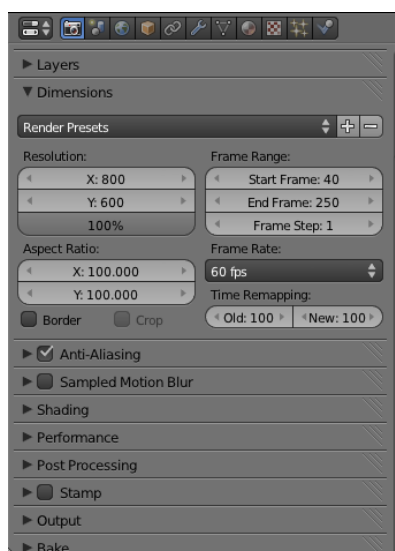
5.1 Frames per second (fps) setting

The frames per second (fps) setting can be set in the render menu:

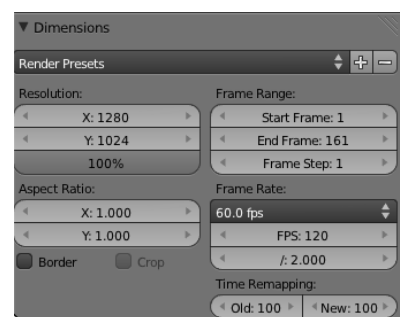


(If Blender game selected on top menu bar)

or



(If Blender render selected on top menu bar)



(If Blender render selected on top menu bar. Here it's $120 / 2 = 60$ fps which is taken into account. The divider (the framerate base) can be < 1)

5.2 Frame rate Multiplier

It allows having a frame rate higher than the fps value using parameter `Frame rate Multiplier`. The frame rate is computed as "`Frame rate Multiplier * fps value`" (see 5.1 for fps setting). The frame rate multiplier is limited to 10. If the computed frame rate is higher than 10, a message is displayed in the log file and `Frame rate Multiplier` is reset to 1 for the export.

For example, if the file is set with:

`Frame rate Multiplier = 3`

With `fps = 30`, the frame rate for the animation will be 90. (This value can be found in the log file at line `SampleRate` in section `IAfHeaders`)

5.3 Remove last frame

The IA file describes the positions of the IA elements through time. For a cyclic animation, the last position is identical to the first.

Eventually, removing this last position in the IA file, may make a cyclic animation smoother.

If Remove last frame is ticked in *IA export config*: last position is not exported, else it is written without any change.

5.4 Trim Animation

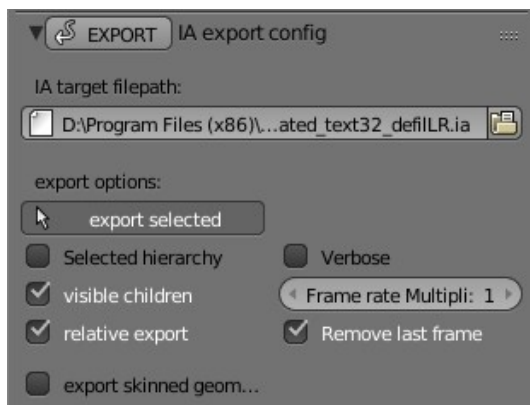
Moments where no actual animation occurs, at the end and beginning of the array of selected frames for exporting, are trimmed out. This leads to a shorter file. This is useful when animating cabs and trains with different animation length not to have to move the start frame and end frame by hand for each set of objects to export.

5.5 Relative export

This setting is enabled by default. When it is active, only animation relative to Blender object's parent are exported. This is useful for animating buttons mounted on moving parts in cab for example.

Unticking this box exports animations relative to the scene for every object which doesn't have his parent included in exported objects set.

5.6 Selected hierarchy



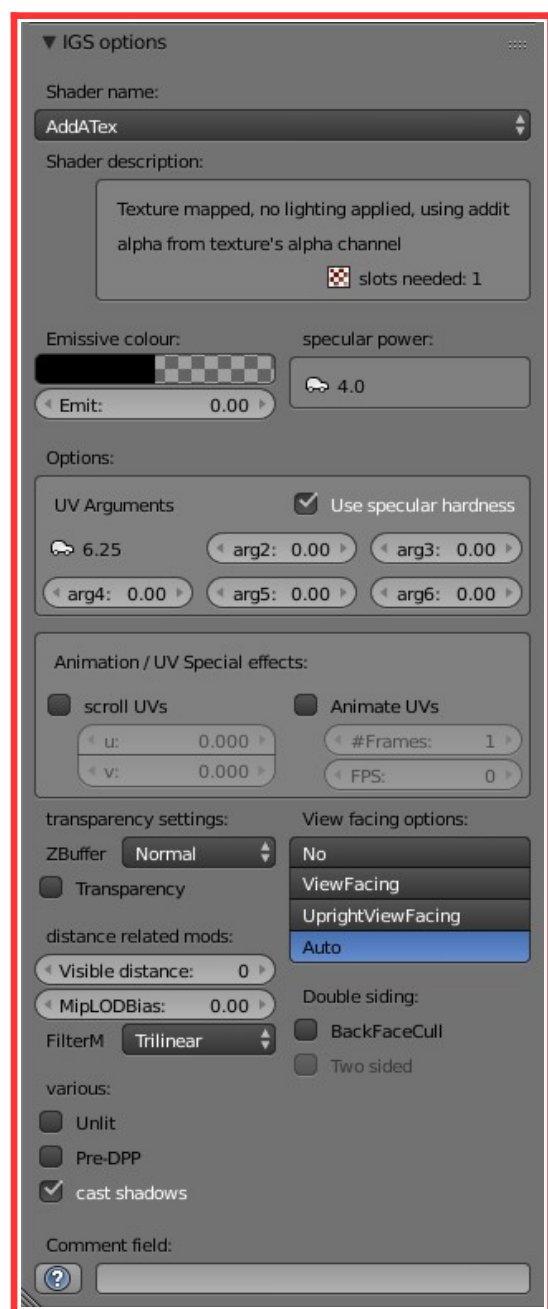
This option is only visible in *export selected* mode.

When active, this will export children of selected pose bones and objects recursively.

Else, only selected pose bones and objects are exported.

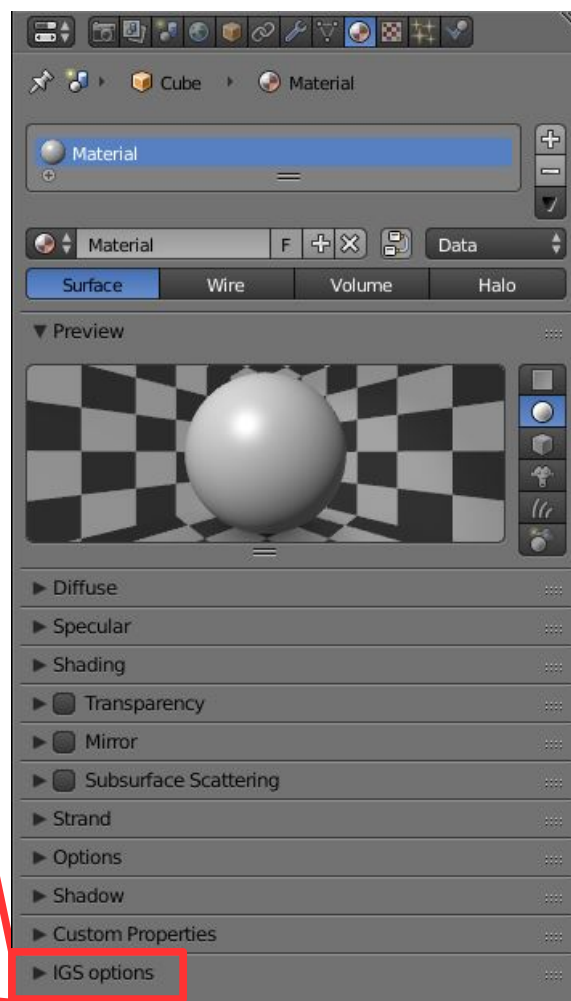
6 Material configuration for IGS Export (IGSE)

Material panel IGS options provide powerful tools and UI for setting properties within Blender.



It is located at the bottom of Properties > Material tab and is visible when Blender Game or Blender Render engines are in use.

Left is an overview of its default settings. These will be explain further later in this document.



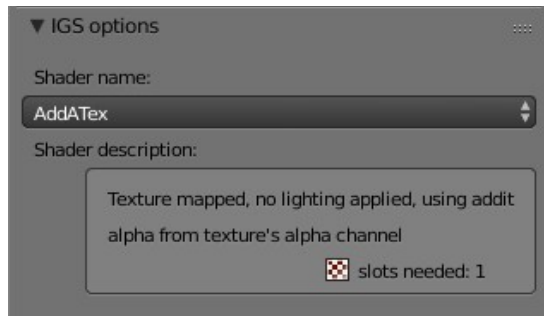
Some options might be greyed out depending on the shader used because all features are not available for all shaders.

Features will be described here from top to bottom.

6.1 Shader assignment

Shaders can be assigned to a material by 3 different manners :

6.1.1 Using UI shader list



Below the `shader name` label is a menu. Clicking on it unfolds a huge pop-up where you can select your shader name. Selecting one will update the `shader description` box with appropriate informations about use and expected number of texture slots. Failing to use the right number of texture slots for a registered shader will generate an error on export.

6.1.2 Using a custom shader



Selecting “\custom shader\” in shader list brings up a new field where a non-registered shader name can be used. It also supports usual and short shader name but it is not intended to be filled with theses!

Example: `TrainBump.fx` is a non registered yet working 2 slot shader 1st for diffuse, 2nd for normal map. It can only be set by `custom shader name` field (or *legacy method* – see 7.1).

6.1.3 Using Legacy method

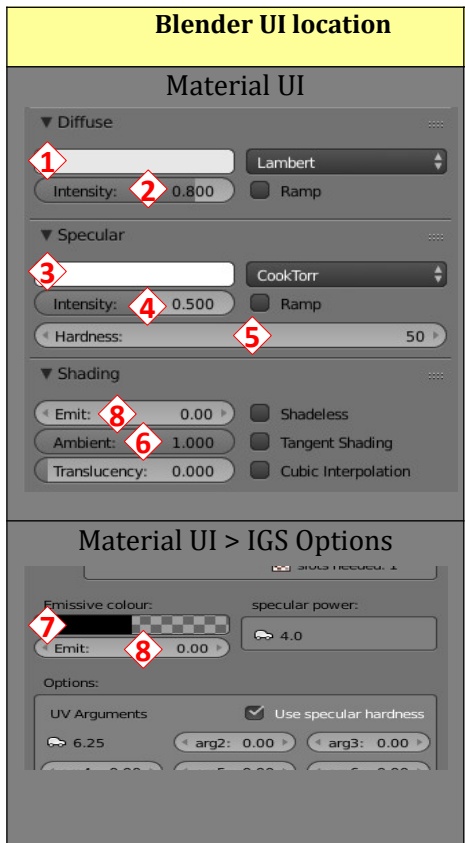
The first enabled texture slot is still checked for legacy compatibility if texture type is `None`. It was used for shader name, whether it is the full name or the short one.

Legacy shader name mode has priority over other shader name selection techniques on export.

Slot order demonstration on the table below (7.1)

6.2 Ambient, diffuse, specular, emissive

In the material menu, the following panels are used for these settings:

Blender UI location	Blender field
 The image shows two Blender Material UI panels. The top panel is 'Material UI' with sections for Diffuse, Specular, and Shading. The bottom panel is 'Material UI > IGS Options' with sections for Emissive colour, specular power, and UV Arguments. Red numbered callouts (1-8) point to specific fields: 1. Diffuse colour, 2. Diffuse alpha, 3. Specular colour, 4. Specular Intensity, 5. Specular Hardness, 6. Ambient, 7. Emissive colour, 8. Emissive strength.	1. Diffuse colour 2. Diffuse alpha 3. Specular colour 4. Specular Intensity * 8 → Specular power can be previewed through the specular power box. 5. Specular Hardness (fidgeted) → Phong size can be previewed in place of UV arg1 when use specular hardness is ticked. See 6.3 - UV Arguments 6. Ambient 7. Emissive colour + alpha 8. Emissive strength

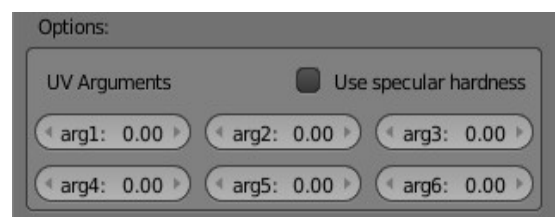
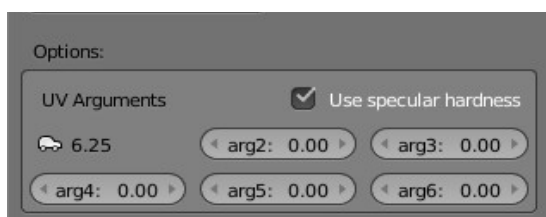
6.3 UV Arguments

There are 6 UV args (short for “UV Arguments”) which have various effects depending on the shader used. But the first one is often used for the same setting:

arg1 : “Phong” size (values between 0<64)

This is the size of the specular highlight: this is close to blender material's “Hardness” visual effects.

When “Use specular hardness” is enabled, Blender's specular hardness is translated to TS hardness system then affected to arg1 on export. Else, arg1 can be modified freely.



IGSE prints this to logfile for each material:

```
BRIAGE:: 'PROPERTY': UVArguments for material xxxx: 0.0 0.0 0.0 0.0 0.0 0.0
```

Here are some examples of their functions for different shaders (from <http://the-art-of-rws.blogspot.fr>):

- TrainBumpSpec.fx
arg1 : “Phong” size (values between 0<64)
- TrainGlass.fx
arg1 : “Phong” size (values between 0<64)
arg2 : Intensity of the reflection (day-time)
arg3 : Intensity of the reflection (night-time).
- WaterCubeMap.fx
arg1 : “Phong” size (values between 0<64)
arg2 : 'Wobble' factor (values between 0<1)
arg3 : Movement speed (0.02 is a good value)
arg4 : Oscillation height (0.02 is a good value)
arg5 : Oscillation scale (6 is a good value)

6.4 UV Special effects

These are special behaviours available for some materials. They allow some sort of texture animation. You can activate them with their associated radio-buttons.



Scroll UVs is available for non-fx shaders. When enabled, U and V coordinates of material's 1st texture slot are scrolled respectively left and down of "image x*u value" and "Y dimension*image y" each frame.

In other words, UV scroll function scrolls

normalized (between 0 and 1) u and v coordinates each render frame.

Animate UVs allows use of "animated" textures (more like a sequence of different images). See this for more informations:

Here is a purpose **modified** version of [the article from Railsimilarity](#):

Train Simulator 20XX supports animated textures. Animated textures refer to a series of identically sized frames or textures which the game will load and display frame-by-frame. Some examples of animated textures would be :

- traffic light sequence
- flashing lights on a police car

The process is fairly straight-forward. The engine relies of the fact that the frames are named sequentially, and once the material is set up to animate, it will loop through these sequentially named frames at a given speed.

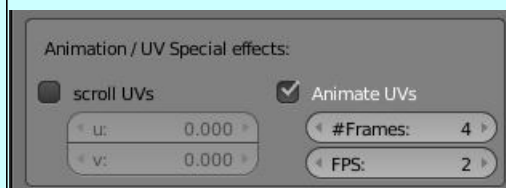
Here, in this example, we will setup a simple 4 frame animation.

1. Initially create the sequence of 4 frames and name them as follows:

XXXXX_anim1.dds, XXXXX_anim2.dds, XXXXX_anim3.dds, XXXXX_anim4.dds etc (where XXXXX is the texture name).

2. In Blender, assign a simple shader such as `TexDiff` (or `AddATex` for glows) to the relevant polygon or mesh. Then in the material, point the texture slot to the first texture XXXXX_anim1.dds. It's important to point to the first texture so the game engine knows where to start the sequence.

3. Still in the material tab, scroll down to find the "UV Special Effects" area, and tick the "Animate UVs" button. Also insert the correct number of frames for your animation in the area marked '#Frames' (in our example this would be 4). Also choose a frame rate in the 'FPS' field. In our example we'll choose 2.

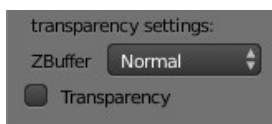


4. Export the shape.

NOTE : The export process will only compile the textures that are referenced in the scene IGS file. As we only reference the starting frame (XXXXX_anim1.dds) in our IGS, only this single frame of the animation will compile. If we view this in the game, we will see the first frame OK but the other frames will be represented as "MISSING TEXTURES". There is a simple solution to this where we can simply force the missing textures to compile from within the blueprint editor. To do this, navigate to find your textures in the blueprint editor and one by one right-click on each one and choose the 'export' option from the roll-out. Once all the textures have been compiled, the shape will be ready to be viewed in-game.

Freely adapted for BRIAGE (2016) from Railsimilarity (2009)

6.5 Transparency



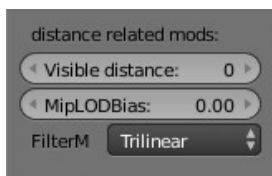
Transparency settings allow use of transparency related parameters with this material.

ZBufferMode allows modifications of how TS direct3D engine computes Z depth when encountering this material in relation with the shader selected.

To get 8-bit alpha with non-fx shader set this to “test only”

The Transparency checkbox is the former AlphaTestMode toggle. It enables transparency for fx shaders, mostly 1-bit alpha.

6.6 Distance related modifiers



These interface material properties with in-game camera distance.

Visible distance when set at another value than 0 sets maximum distance in meters where this material is viewable. Else, it's considered infinite.

MipLODBias pulls MIPmaps change distance closer(negative values) are away(positive values). This is a particularly effective way of fixing with distance-blurred textures in TS by applying a negative value here.

FilterMode sets the filtering mode used for the texture in-game.

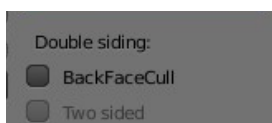
6.7 Viewer-facing options



Viewfacing enumerator allows you to choose the viewer facing mode you want to apply to your material.

- ➔ No mode is self-explanatory.
- ➔ ViewFacing mode enables all-axis view-facing rotations
- ➔ UprightViewFacing mode enables local Z-constrained view-facing rotations
- ➔ Auto mode applies the best or most common setting corresponding to the selected shader name.

6.8 Double siding



These two boxes allow culling faces' back for non-fx shader and double-siding for fx shaders.

6.9 Various settings



Unlit activate LMKeepVertexColors flag. Fun fact, it doesn't make your material unlit: please use specific shaders for this purpose.

Pre-DPP is a comment field toggle (see 6.10) added on export to the comment field. (DPP means **Deferred Post Processing**, these are effects applied after rendering objects like antialiasing or other 2D effects.)

Cast shadows is taken from Blender's interface and activates LMCastShadows flag.

6.10 Comment field



This field allows manually entering comment-driven material behaviours. The ? button brings a pop-up with a list of known comment field toggles.

Here is copy of the list:

```
:forceprelit      :fpsfraction      :forceunder
:pointlit         :fadedist         :fullalphasort
:pre-dpp          :noclip           :occlude
:unfog            :startinvisible   :alternode
:animdelay        :startnocollide   :shatter
:animmaxframe     :nocompress       :objectstick
```

6.11 Other properties

Some properties are scattered across Blender Properties area :

zBias : Material tab > Options panel> Z Offset

IGobject_color : (only set by IGS's main objects) Object tab > Display panel> Object color

7 Materials textures settings

Textures are essential to most of available shaders; don't forget to set them accordingly to what is required by the selected shader.

7.1 Texture slots mapping with TS20xx shaders

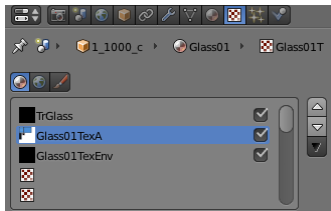
IGSE texture slot available dispositions :

n-th Activated slot	Standard	Legacy	TrainGlassWeatherEffects.fx
Slot 1	Texture 1	Shader name (6.1.3)	Texture 1
Slot 2	Texture 2	Texture 1	1 Texture 3
Slot 3	Texture 3	Texture 2	Texture 4
Slot 4	Texture 4	Texture 3	
Slot x	Texture x	Texture x	

(1) Texture 2 is automatically processed by the exporter.

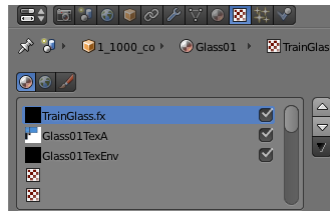
Some shaders usage examples are given in annex 10.2 and a non-exhaustive shaders list is given in annex 10.1.

Example of slots setting:



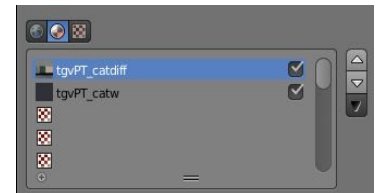
using short name:
TrGlass

or



using long name:
TrainGlass.fx

or



using shader list or custom shader
field:
TrainGlass.fx

The slots must be enabled (☒) to be taken into account. The enabled slots don't need to be contiguous. Only the enabled slots order matters.

7.2 Mapping panel for a texture slot associated with an image

Typical setting of the Mapping panel for a texture slot associated with an image:

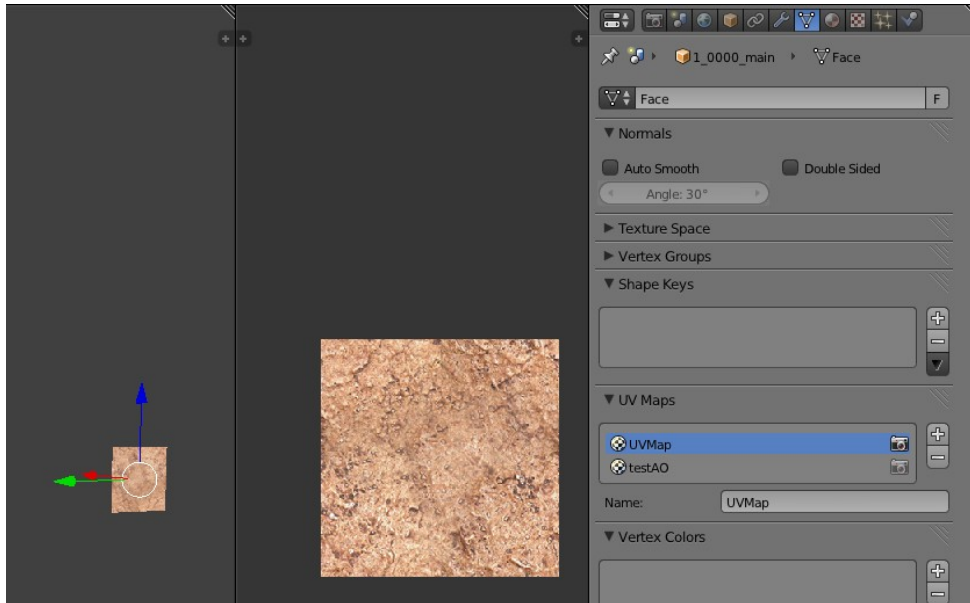


The "Map:" field may be left empty when using object active UV channel for mapping this texture slot.

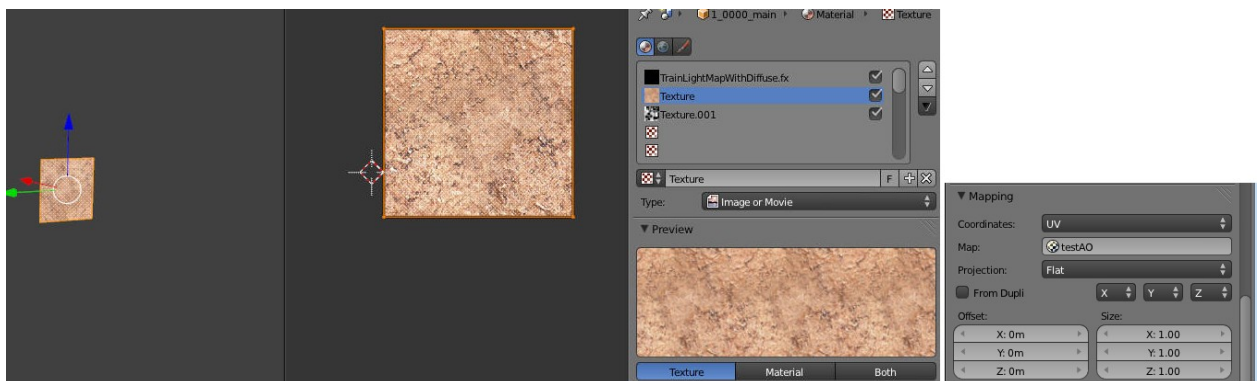
Different mapping informations may be used for different texture slots of the same material.

Here is such an example with shader `TrainLightMapWithDiffuse.fx` :

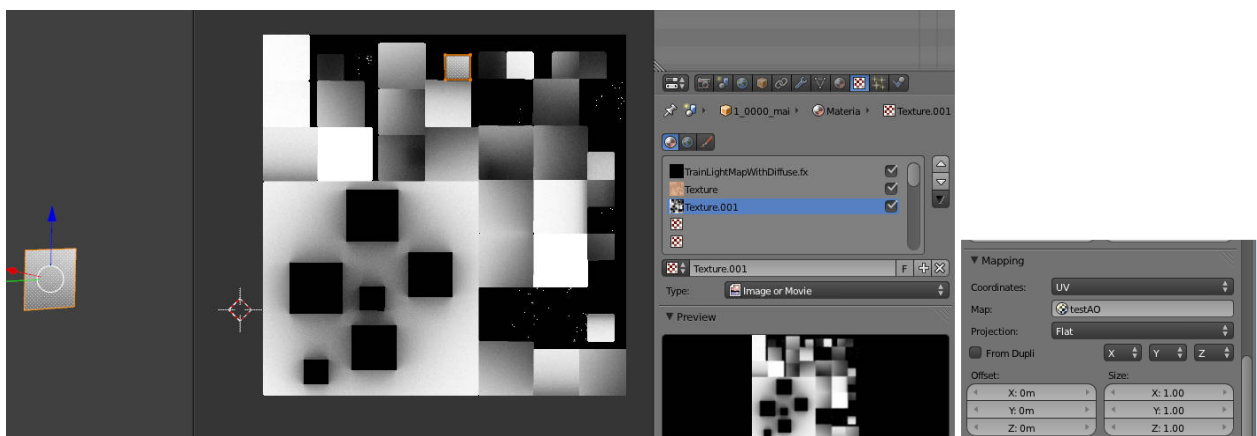
2 UV maps are defined on this object:



One for the main slot:

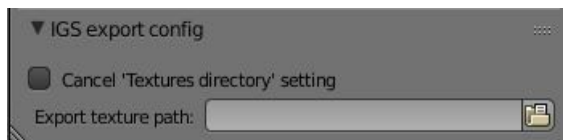


... and one for the second slot ("lightmap" slot of `TrainLightMapWithDiffuse.fx`) :



7.3 Individual texture's path mapping options

There are different ways of binding textures path to materials on export.



Cancel 'Textures directory' setting prevents IGSE from using 'Textures directory' (see 4.5) setting on this texture.

>> If using “*New texture mode*”: wherever you export your IGS, its texture component will point to the file used by Blender.

>> If using “*Legacy texture mode*”: the same relative filepath from blend to texture will be applied to IGS. For this setting to work properly, work in your source folder sub-directories where you will export your IGS to, or make sure your working directories mimic exactly end-directories hierarchy.

Export texture path precise a texture path to use on IGS export instead of Blender's one. This field cancels 'Textures directory' setting too. To only change texture's name: please use 'texture strings to replace' table(see 4.5). Option is disabled if field is empty.

This setting is particularly useful when reskinning an “unreskinable” assets (like lofts) or when textures are not available on export path.

>> If using “*New texture mode*”: wherever you export your IGS, its texture component will point to the file specified by Blender.

>> If using “*Legacy texture mode*”: the same relative filepath from blend to texture will be applied to IGS. For this setting to work properly, work in your source folder sub-directories where you will export your IGS to, or make sure your working directories mimic exactly end-directories hierarchy.

8 Notes

Here are various notes you should take into account while using BRIAGE.

8.1 Using Duplication

Duplis are objects or groups of objects instanced in Blender by Duplication panel of objects.

These instances are now exported by BRIAGE with the following strategies.



8.1.1 DupliFrames

These are merged with duplicator's on export.

8.1.2 DupliVerts and DupliFaces

These are merged with their respective original object on export.

8.1.3 DupliGroups

The content of the groups is *parented to the duplicator* (mesh or empty) and *renamed with duplicator's name*.

Using an empty without parent as duplicator preserves group's hierarchy; else, group objects with no parent are parented to the duplicator.

The duplicator cannot be a snap point. Duplicator's name may or may not use TS naming conventions, its detected **name is inserted at the beginning** of object's and bone's names of the group instances. You must *be careful not to exceed the overall 24 chars* limit after insertion, else name will be truncated. If a **"*" is present as last** char of duplicator's name, name will be **appended to instance's names** without "*".

Dupli animations can be exported as well.

8.2 Using Snap Points

Export of Snap Points is now available.

They obey to 3DSmax naming rules; see '*TS directory*'/*dev/Docs/TS2015Tech.pdf*, chapter 4. Here is a **modified** extract from the above-mentioned documentation:

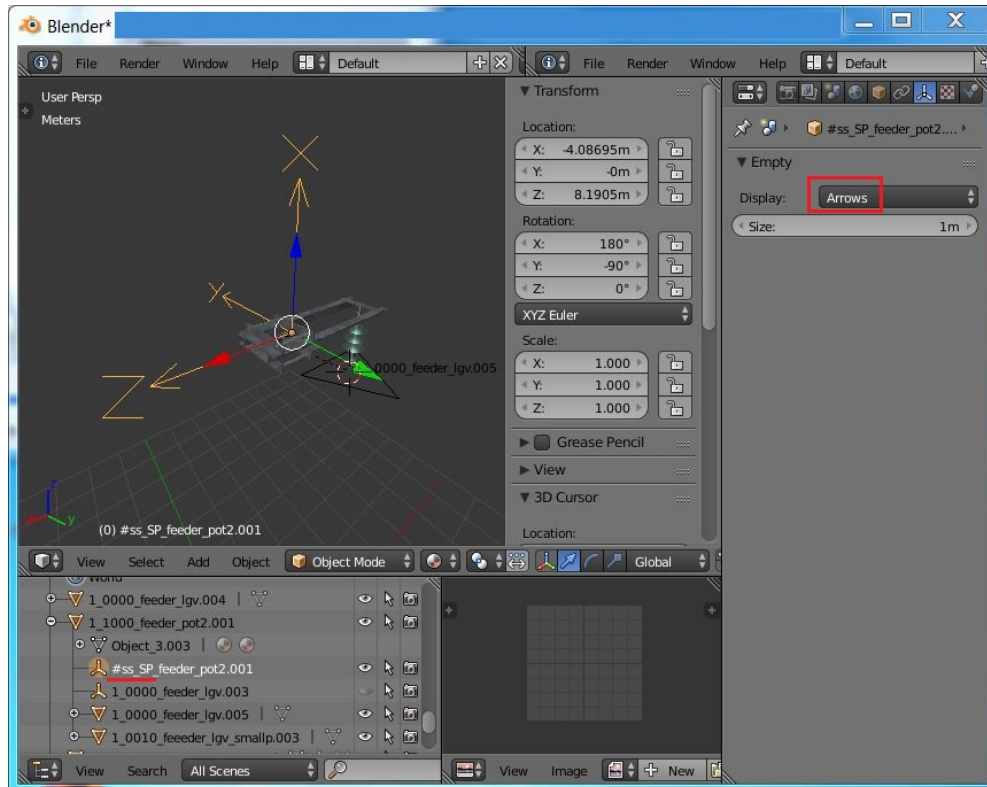
1. Place an “arrows” type empty in the Blender scene. The naming convention is important so the blueprint editor can recognise this as a snap point.
 - **#ss_**xxx is a **source** snap point
 - **#st_**xxx is a **target** snap point
 - **#sb_**xxx is both a **source** and **target** snap point
 where xxx is a **unique** name, in this case SnapOut2 (I was using SnapIn for target snap points)
2. Orient the snap point object so that the Z axis is pointing in the direction of the desired alignment when snapped to a target. In general, for all snap point types, this is usually away from the model's surface
3. Setup remaining snap points.
4. All done! Now export as IGS.
5. Add as the model of a scenery blueprint, or any other blueprint that has a container component.
6. Hit export. The snap points will automatically be added to the container component on export, and by default these will be the
 Kuju\RailSimulatorCore\Editor\Gizmo\DefaultScenerySnapPoint*.xml blueprints.

NOTE: You can create your own snap point blueprints and configure various settings for these. To use your own blueprints, you will have to manually add the snap points with the names you gave them in Blender as children of the blueprint, e.g. **#ss_SnapOut2** would be named **SnapOut2**. You assign your custom snap point blueprint as the blueprint ID, and hit export. The blueprint editor will match the names of the snap points in the model to the ones in the blueprint and override their matrices automatically.

Freely adapted for BRIAGE (2016) from TS2015Tech.pdf (2015)

In Blender they follow these additional rules:

- **Place an "arrows" type empty (Display type)**
- They can have a parent object
- They mustn't have any children



Take a look at "SampleWithDots.blend" example file for a demonstration.

8.3 Using skinning and bones

Export of armatures and skinning is now available.

Armatures are exported when using Export skinned geometry function (see 4.2). Corresponding Vertex groups weights are exported when an armature modifier is present on a skinned object. *Envelope deformations are not supported* yet if not applied to Vertex groups. Skinned objects must only use "Skin...fx" shaders to work properly. Skinned objects doesn't need to be child of an armature they only need an armature modifier set to **deform following vertex groups weights**. Vertex groups names must be the same as the concerned bone name. It is recommended to take a look here: Blender manual "[Weight edit](#)", "[Skinning to Shapes](#)" and more generally to "[Skinning](#)" chapter.

All objects within an IGS doesn't have to be skinned. Used armatures must be exported (it follows rules defined by selected Export mode). Make sure all your **bones have unique names even across armatures!**

You might want to disable this feature for armature-based animations which doesn't need any deformations on any object to speed up performances.

Take a look at "Prez.blend" example file for a demonstration.

8.4 About model export

8.4.1 General notes

The use of negative scaling requires applying Ctrl A “scale”.

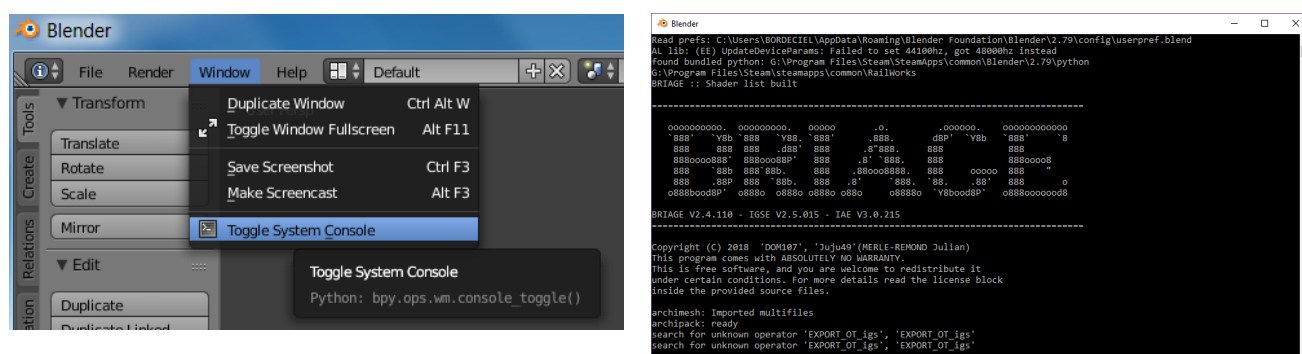
"current frame" is set to 0 during the export. The different animated objects must be in a **rest position at "current frame" = 0**. At the end of the export, "current frame" is reset to the value before performing the export.

Only 256 objects can be exported within an IGS including LoDs and without snap points.

Max number of faces and vertices is unknown.

8.4.2 Reading logs

A log file is not printed if not especially asked with the “Verbose” toggle. But a shortened log is available after each export in Blender's system console here:



You will note that when BRIAGE starts properly, it prints this in Blender system console: (right picture)

Paragraph 9 lists typical export errors or problems.

Even without any error, check specifically for:

- Messages about name substitutions, could it be:
→ incorrectly named objects.
→ replaced object names
- The objects groups list starting with “----- Groups list -----”.
It can be checked that objects are grouped as expected.
- Some material parameters taken correctly into account(UVArguments, etc...)

The short log file allows displaying the following information:

- **Materials list** with associated shader, first texture and UVArguments parameters.
- The **texture files list**

8.5 About animation

8.5.1 Blender 3D view animation

The exporter processes active animations at the time of the export file saving.

Thus, if, for example, a constraint is disabled in the open model, it will not be in the exported animation. In short, the animation working in the Blender 3D view should be the one played in TS20XX, provided all the objects selection is correct (“Export selection” mode) or all the objects needed for the animation are

visible ("Export visible" mode). Some options allow more control over what should come out in TS, see 5.5, 5.6.

Don't forget to set the frames per second (fps) and the start and end keyframes according to the relevant objects: a switch animation is shorter than a pantograph animation.

Immobile objects are timed from export anyway.

8.5.2 Scaling

Data in exported IA format are positions and rotations. There is therefore no scale. Remember to apply scale to objects **CTRL+A "scale"** to avoid strange surprises. Thus the effect of a "copy scale" constraint won't show up in the TS20XX exported animation. Although, a scale animation could be faked using an armature and skinning vertices, one bone for each stretching direction.

A model and tutorial on "*animating the valve-gear of a steam locomotive*" by Mike Adams can be found on UKtrainsim download section. Though this tutorial is being updated for this export tool, it gives a good example of what can handle the exporter: lattices, bones, armatures, constraints of type copy location / copy rotation, limit rotation.

8.5.3 unrelated animations

It is likely that one Blender file will have several unrelated animations (such as for example, wipers and doors). These different animations can all start at keyframe 0. However when animation is played in Blender all the animations in a specified keyframes range will animate. This is no problem for the export as, once the keyframes range is set, you'll only select objects to export animations for (in this example, the wipers) and the other moving objects will not be included.

If you prefer independent animations in the Blender file, you can use different keyframes ranges. Select your object(s) and its(their) related parent objects and set the keyframes range to when they animate and finally export as IA file.

For example, a wiper animation between keyframes 0 and 100, a door animation between 120 and 200, etc.. To export the door animation, select your doors and their related parent objects and set the keyframes range between 120 and 200.

8.5.4 180° animations

Animating needles in cab doing more than a 180° rotation can be challenging due to stuttering. It is advised to use a Frame rate Multiplier of 7 and have an animation with an odd number of frames. Devdocs point out a length of 25 frames.

8.5.5 other IA tips

The dope sheet or the graph editor allows to easily move keyframes if necessary.

It is not necessary to select all objects of an animation set and especially an object with an already selected parent if using "selection mode".

Among the selected objects ("selection mode" enabled) or the objects below the scene object ("visible mode" enabled), the criteria of inclusion in the animation export are (see example at Annex 10.5):

- ◆ They have a constraint or an animation.
- ◆ They are direct parent of an armature.
- ◆ They have a child (parentage created with CTRL+p) with a constraint or an animation.

When enabling "selection mode", **be sure that the selection is consistent with the IGS** structure or bring corrections with Relative export and Selected hierarchy options. For example, if the IGS file was created exporting the entire model, the selected objects for animation should be at the level under the scene object (otherwise, risk of unwanted translation when playing the animation without using previously cited options for adjustments).

By default, the log file lists the selected objects and those taken into account in the export.

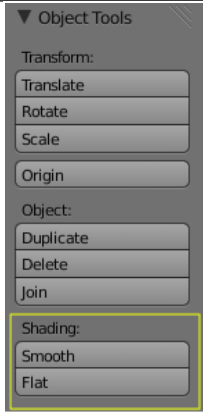
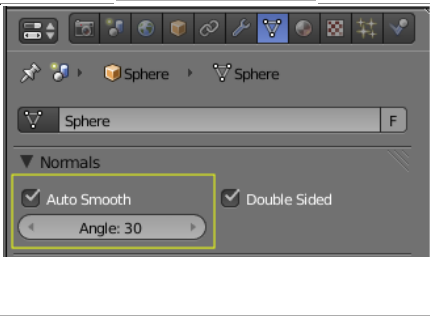
8.6 About smooth vertex normals

The exporter takes into account whether to render the object fully smoothed, flat or **auto-smoothed** meshes. It also reads and export **custom normals** and **sharp edges**.

You can read details on smoothing here:

<http://wiki.blender.org/index.php/Doc:2.6/Manual/Modeling/Meshes/Smoothing>

I use below some information from this link for a few explanations.

The easiest way is to set an entire object as smooth or faceted by selecting a mesh object, and in Object mode <u>click Smooth in the Tool Shelf</u>. This button does not stay pressed; it forces the assignment of the "smoothing" attribute to each face in the selected mesh. Click the <u><i>Flat</i></u> button in the Tool Shelf's Shading panel to cancel the smoothing effect.	
Auto smoothing can be enabled in the mesh's panel in the Properties window. Angles on the model that are smaller than the angle specified in the Angle button will be smoothed for the export and during rendering (i.e. <u>not in the 3D view</u>) when that part of the mesh is set to smooth (i.e. <u>Smooth was also clicked</u>). Higher values will produce smoother faces, while the lowest setting will look identical to a mesh that has been set completely solid.	
Another way of changing the impact if smoothing is to use the edge split modifier. But this might just be useless as this is already done internally on export...	

If you want to use custom normals for improved lightning effects and cleaner normal maps, have a look at this Blender Add-on over there: [Y.A.V.N.E. \(Yet Another Vertex Normal Editor\)](#)

8.7 About materials

8.7.1 Performance notice

It is important to keep the number of different materials low because each material contains a lot of informations which occupies large disk space but also RAM when exporting which slows down the process. The number of polygons impacts the game's fps far less than a high number of materials.

8.7.2 TrainBasicObjectDiffuse.fx 1-bit alpha

`TrainBasicObjectDiffuse.fx` and most *.fx shaders can be used with 1 bit transparency in the alpha channel provided it's only black (full transparency) or white (no transparency); shades of grey are . For the transparency to be active, `Transparency` must be enabled in material's IGS options.

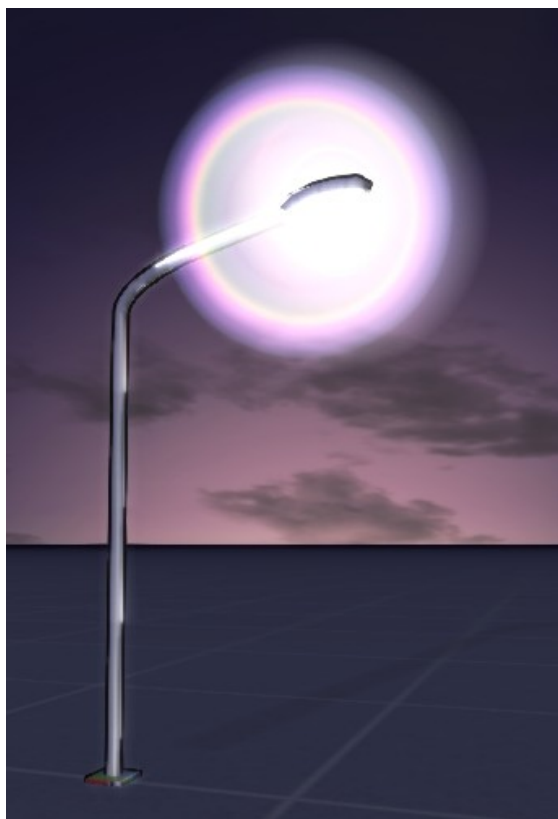
8.7.3 Vertex colours

Active and renderable `Vertex colors` channel is exported.

Be careful when using smaller IGS setting, if vertices have too different colours for their adjacent faces they will not weld correctly together, causing performance issues in TS.

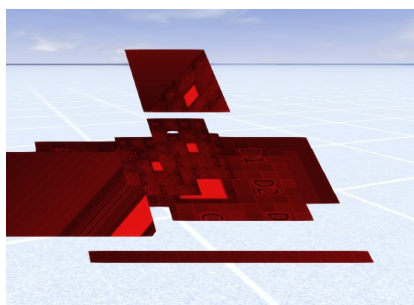
8.7.4 Viewer facing properties

Viewfacing properties are supported for `AddATex` shader.



Take a look (left) at “UglyToonStreetlight.blend” and “SampleWithDots.blend” example files for a demonstration. Light halo only appears at night.

Viewfacing value is automatically set according to shader names if they contain "Upright" and/or "ViewFacing" (e.g.: `TrainUprightViewFacingFlora.fx`) if you set Viewer-facing options to **Auto**.



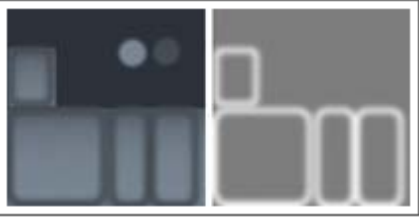
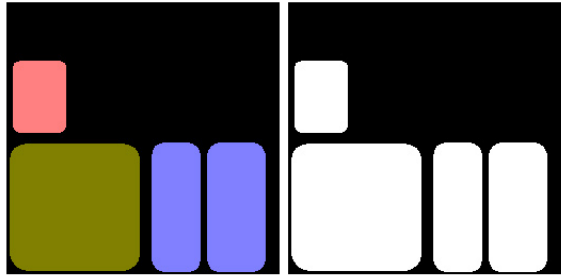
Be careful while using shaders in upright view-facing mode (figure above): quads will turn around local Z axis and vertices will use a matrix defined from main object rotations to position themselves. Be sure your main

object doesn't have some rotations if you use Upright View-facing quads as they will appear deformed: (screenshot on the left)

An invisible object must be added otherwise the object once added in a scenario won't be pickable. One way is to add a cube with a UV map initialized and use the “invisible” shader. The “invisible” shader must not be accompanied by a texture file else, it will generate an error (see example 10.7).

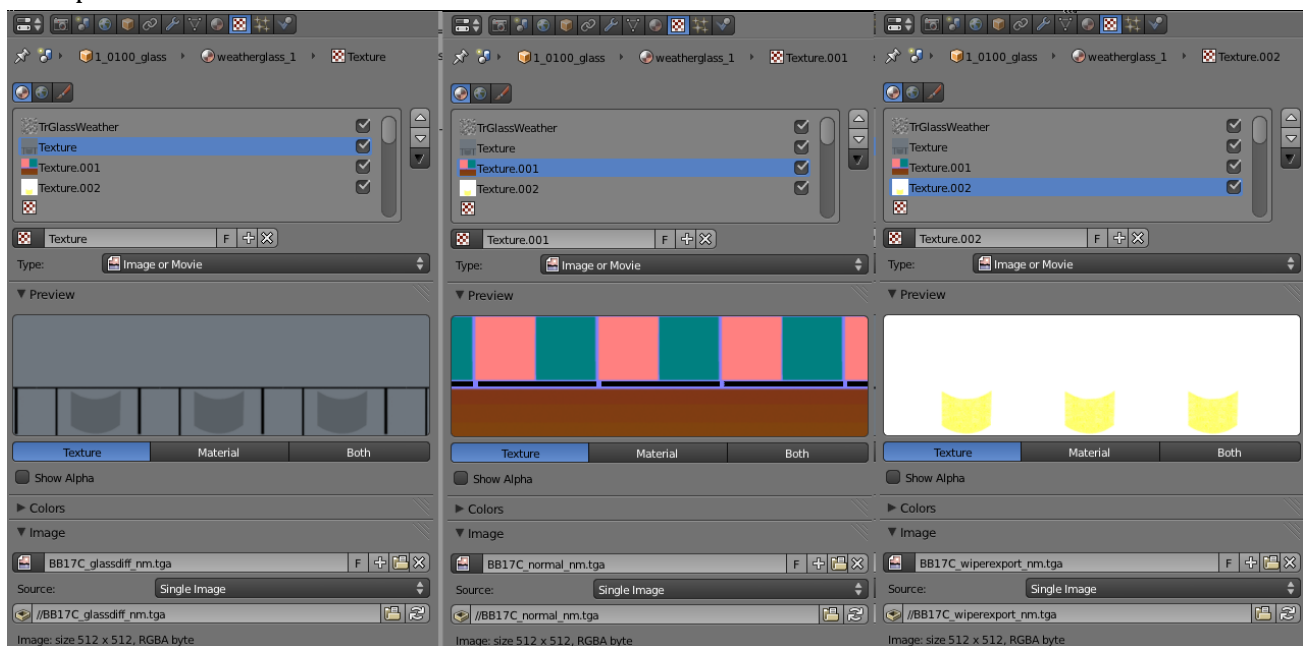
8.7.5 Setting-up TrainGlassWeatherEffects.fx

TrainGlassWeatherEffects.fx is processed by the exporter. **The material name with such shader must be weatherglass 1 then weatherglass 2**, etc... (According to the number of materials using this shader) you only need to have **3 slots** with the useful texture files (The texture filenames is the user choice. The 3 images below are from https://www.christrains.com/en/ts_faq_weatherglass.html) :

windiff_nm.dds : Main texture with alpha channel.	
winnorm_nm.dds : normal map texture to define where the raindrops are.	 R=128 G=128 B=255 FRONT WINDOWS R=128 G=128 B=0 BACK WINDOWS R=0 G=128 B=128 LEFT SIDE WINDOWS R=255 G=128 B=128 RIGHT SIDE WINDOWS Diffuse (left) and alpha (middle) channels for slot 3. RGB Colours (right) for window directions
winwmotion_nm.dds : Texture to define the wipe out pattern of raindrops. May be generated using "winmotion.exe" in TS directory	

For TrainGlassWeatherEffects.fx, the exporter automatically adds an additional slot for TS20XX. (dummy cubemap slot) UV arguments are not enabled for this material.

Example in Blender:



See also some shader usages in annex 10.2.



In-game result (left),
CD 460 by *4a 61 63*
68 79 48 6d

9 Typical export errors or problems

Most of export errors are caught on export and display pop-ups about what the trouble is. Caution: in Blender, pop-ups disappear when they are no more under the mouse cursor but BRIAGE writes then to Blender's System console(Info area > Window > System console) and/or log file for the user to be able to find them again.

9.1 Warnings

9.1.1 About Asset Editor cache

When exporting the **ia animation** trying different parameters, it's better to exit the Asset Editor, sometimes clean the cache and **restart the Asset Editor**. Otherwise, the Asset Editor might not update the animation properly.

9.1.2 About Asset Editor preview mode

While trying different parameters with **lofts**, modifications does not update accordingly. Please close and **restart the preview**.

9.1.3 About TS live object export

Exporting objects **from Asset Editor** to an enabled provider and product while TS Route Editor is open on your computer and then reload (not F9 but deactivate then reactivate its checkbox) the product **can and will corrupt your route on save and then crash TS**. Never do or try this on any valuable route!

9.2 No mesh or empty object found export. Export Stopped



This error is generated when nothing is found which can be exported. To solve this :

- if in "Export selected" mode: select objects.
- if in "Export visible" mode: ensure all the objects you want to export are visible in the 3D View.

9.3 Please export to the same drive your data are from! Texture 'xx' not added.



This appears when you export a file to another drive your textures are from, the best way to avoid this is by using TS's source folders as working directories while developing your model.

It also appears when you are exporting an object from a scene never saved before and Blender executable is not on the drive you are exporting to.

Example: Your textures are under **D:/myWork/myTrain/myTexures/** and you export to **C:/steamLib/common/Railworks/Source/myProvider/myProduct/RailVehicles/myTrain/** the error will

pop-up. This will not happen if textures are under **C:/myWork/myTrain/myTexures/** but textures paths in the igs will point there and your model will never work on another computer than yours.

To fix this behaviour (picking one is enough):

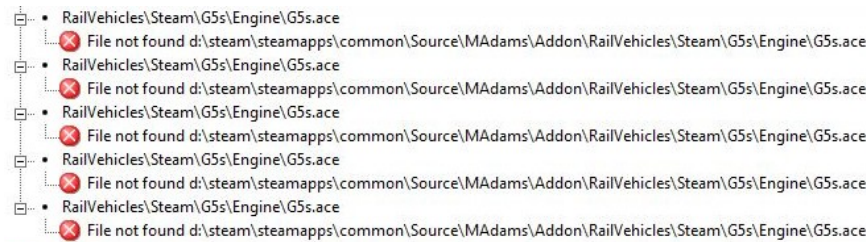
- If you are accustomed to the old texture processing way, uncheck this and it should work as before:



- Use provided texture path 'on-export' modifier tools (see 4.5 and 7.3)
- Use TS's source directories as working directories

9.4 No texture files found in asset editor or blueprint editor

If no texture files are found when opening the xml model file in the asset editor or the blueprint editor



Check in the log file that the export used the proper texture mapping informations. If you are not familiar with the new BRIAGE texture workflow, uncheck “write from *.IGS destination”:



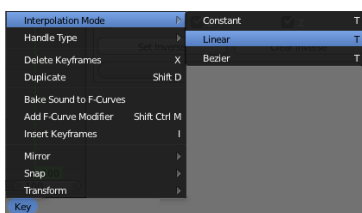
9.5 The animation is working properly in Blender but not in the asset editor

If nothing moves, check in the main blueprint “Anim Set” section that all the animation sets have “Animate in editor” set to “True”.

If the parts are not moving as expected:

- **Check the igs / ia export consistency.** For example, selected objects for animation are not under the scene level whereas the igs export was made for the entire model (no selection).
- **Check in the igs export log file (“Groups list” section) that a moving part was not merged with another object.** If this is the case, add keywords in the Custom Keywords table, if the model has not too many objects, disable hierachy processing.
- **Check the FPS value** (number of frames per second) in the igs exporter log file. This value is listed in every description of a material even if option "verbose" is not checked. Setting-up FPS is explained in paragraph 5.1.
- Exit the Asset Editor, clean the cache and restart the Asset Editor.

9.6 The animation is stuttering



If you can't get a smooth animation, try “Key” menu -> “Interpolation Mode” -> “Linear”, either in the dope sheet or in the graph editor.

You can also try to lower the framerate divider (framerate base: see 5.1) or try to use framerate multiplier option: see paragraph 5.

9.7 Image filename contains dots '.' which may cause unexpected results in material 'xx', in texture slot 'yy'

This warning appears when the texture filename contains more than 1 dot because the blueprint editor will process these thinking the last part after the first dot he sees is a file extension. Therefore a “.safe” is appended at export in this case.

9.8 IGSE configuration parameters are not working as expected

Check the igs log file in Blender's system console or by performing an export with **Verbose** option activated and check the output:

- Check all your exported objects have their scale applied (active object > CTRL+A > “scale”). This is known to cause untraceable bugs during export.
- Then, check its contents: Are the expected IGS parameters listed as expected (shader names, texture names, etc...)? If not refer back to sections from this very manual about setting correctly the parameters which failed to export.
- For transparency, you may need to change some parameters under **Material IGS options panel** > **Transparency settings** according to the shader used (see paragraph 10.2).
- If you still encounter problems, please contact the author or current maintainer(s) of BRIAGE about your issue with a detailed log file and eventually your *.blend file.

9.9 An object of an animated set is not in the expected rest position

An object of an animated set may not be in the expected position before running the animation. Still, when running the animation and when the set returns to the rest position, all objects of the animated set are well positioned.

To fix this problem:

- If you still encounter problems, please contact the author or current maintainer(s) of BRIAGE about your issue with a detailed log file and eventually your *.blend file.
- Check the object position with current frame set to 0 (the igs export is done with "`current frame`" = 0).
- Before animating an object, ensure to use Ctrl+A “rotation & scale”.
- If the badly positioned object is a child, clear the parent link (**Alt+P** Clear and Keep Transformation). Add the link with the parent using the **ChildOf** constraint + **Set Inverse**. Don't forget to export again the igs file. The object must then be included in the selection when exporting the animation.

9.10 "Inconstent LODing" error or some objects are offset

Check that:

- An object of LOD level N has a parent of LOD N-1 (parentage created by Ctrl p).
- The child name must be the same as the parent name, apart from the LOD level and distance
- The child LOD distance must be > than the parent LOD distance
- Some objects with different names haven't been truncated to the same 24 chars name to comply with LOD naming conventions

10 Annexes

10.1 Shaders list

This list is exhaustive: other shaders are used directly by game engine or unfinished, contain bugs or just less useful copies of other listed shaders in the use which can be made of them from an 3D artist point of view.

This table is meant to give informations about which slot each texture type should use. Hence, shaders which don't have their source file(s) included with TS distribution cannot be listed.

Cubemaps can be replaced by dummy textures, TS always uses its default cubemap instead.

All *.fx shaders support at least 1-bit alpha transparency mode when Material IGS option Transparency is ticked. Even if “RGBx” is specified. *I don't know any exception to this rule*

Images contains usually 4 components : RGBA (3 colours [RGB], 1 transparency [A]). Unused components are replaced with an “x”. By default, “RGBA” designate here 8-bit per channel textures used as diffuse maps with a transparency channel unless stated otherwise.

“xxxx” represent dummy textures: none of their channels will be used whatever they contain. It is advisable to keep these as small as possible (e.g.: 8x8).

Shaders table colour code	ConvertToGeo.exe known supported shaders (App.fbk list)	Blender / 3DsMax UI listed shaders	Shaders I wanted to add
ConvertToGeo.exe known supported shaders (App.fbk list)	ShaderName	ShaderName	ShaderName
Blender / 3DsMax UI listed shaders	ShaderName	ShaderName	ShaderName
Shaders I wanted to add	ShaderName	ShaderName	ShaderName

Some **Blue listed** shaders might bug *ConvertToGeo.exe* while exporting producing “invalid parameters” errors or provoke an “infinite export” bug.

Each Grey listed shader have been tested and is working properly in TS.

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
AddAlphaDiff	No texture, additive vertex alpha with diffuse colour				
AddATex	Texture mapped, no lighting applied, using additive alpha from texture's alpha channel	RGBA additively drawn			
AddATexAlphaDiff	Texture mapped, with diffuse colour, using additive alpha from texture's alpha channel combined with vertex alpha	RGBA additively drawn			
AddATexDiff	Texture mapped, with diffuse colour, using additive alpha from texture's alpha channel	RGBA additively drawn			
AddDiffuse	No texture, with diffuse colour, using additive alpha				
AddTex	Texture mapped, no lighting applied, using additive alpha	RGBA additively drawn			
AddTexAlphaDiff	Texture mapped, with diffuse colour, with additive vertex alpha	RGBA additively drawn			
AddTexDiff	Texture mapped, with diffuse colour, using additive alpha	RGBA additively drawn			
BlendAlphaDiff	No texture, vertex alpha blending with diffuse colour				
BlendATex	Texture mapped, no lighting applied, using alpha blending from texture's alpha channel	RGBA			
BlendATexAlphaDiff	Texture mapped, with diffuse colour, using alpha blending from texture's alpha channel combined with vertex alpha	RGBA			
BlendATexDiff	Texture mapped, with diffuse colour, using additive alpha from texture's alpha channel	RGBA additively drawn			
BlendATexDiffTrans	Texture mapped, diffuse colour, alpha blending from texture's alpha channel, pixels with alpha=0 are transparent (e.g. alphaed fences).	RGBA			
BlendTexAlphaDiff	Texture mapped, with diffuse colour, with vertex alpha blending	RGBA			
BridgeSplit	Not drawn. Use to define areas where track crosses over itself.				
Clouds.fx	RailWorks specific shader for sky dome cloud layer	RGBA			
Diffuse	No texture, just diffuse colour				
DiffuseBumpmap	Unknown	RGBx	RGBx Normal	RGBx Cubemap	

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
DualAddATexDiffDestBlend	Dual textured, diffuse colour, first pass additive, and second pass blended alpha with the alpha of the first texture (e.g. puddles).	RGBA additively drawn	RGBA		
DualBlendATexDiffAdd	Dual textured, with diffuse colour, using alpha blending for first pass and additive alpha for second pass	RGBA	RGBA additively drawn		
DualTexDiffAdd	Dual textured, with diffuse colour, using additive alpha for second texture	RGBA	RGBA additively drawn		
DualTexDiffAddWithLightIntens	Add second pass to first pass, brightness of second pass affected by lightmaps if used	RGBA	RGBA additively drawn		
DualTexDiffAddWithoutLightIntens	Add second pass to first pass, brightness of second pass not affected by lightmaps if used	RGBA	RGBA additively drawn		
DualTexDiffInvisibleStencilBlend	Dual textured, with diffuse colour, first pass invisible, second pass alphaed using alpha of first pass texture	xxxA	RGBx		
DualTexDiffStencilAdd	Dual textured, with diffuse colour, using additive alpha for second texture only where first texture has solid alpha	RGBA	RGBA A*slot1 A additively drawn		
DualTexDiffStencilBlend	Dual textured, with diffuse colour, using blended alpha for second texture only where first texture has solid alpha	RGBA	RGBA A*slot1 A		
DualTexDiffTAlpha	Dual textured, with diffuse colour, using second texture's alpha channel to blend between textures	RGBx	RGBA		
DualTexDiffTrans	Dual textured, with diffuse colour, using second texture's transparency	RGBx	RGBA		
DualTexDiffVAlpha	Dual textured, with diffuse colour, using vertex alpha to blend between textures	RGBx	RGBx		
EmbossBumpmap	Bumpmap for Train 2 prototype or something like that. Takes 2 OR 3 RGBA textures.	RGBA	RGBx Normal	(RGBA) optional	
InvertATexDiff	Unknown following naming conventions: Texture mapped, with diffuse colour, using inverted additive alpha from texture's alpha channel	RGBA			
Invisible	Nothing is drawn; use for invisible collision barriers or to be able to select viewer facing trees				

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
LoftBump.fx	RailWorks specific shader for lofts with diffuse and bump channels	RGBx	RGBx Normal		
LoftBumpAlpha.fx	RailWorks specific shader for lofts with diffuse/alpha and bump channels	RGBA	RGBx Normal		
LoftBumpTrans.fx	RailWorks specific shader for lofts with diffuse/ 1-bit alpha and bump channels	RGBA 1-bit alpha	RGBx Normal		
LoftPostDPPTexDiffSpec.fx	RailWorks specific shader for lofts with diffuse channel and fixed specular post DPP stage and shadow mapped <i>Supports UVarg1 see 6.3</i>	RGBA 1-bit alpha			
LoftTexDiff.fx	RailWorks specific shader for lofts with diffuse channels	RGBx			
LoftTexDiffAlpha.fx	RailWorks specific shader for lofts with diffuse channels blended between 2 lofts	RGBA			
LoftTexDiffSpec.fx	RailWorks specific shader for lofts with diffuse channel and fixed specular <i>Supports UVarg1 see 6.3</i>	RGBx			
LoftTexDiffTrans.fx	RailWorks specific shader for lofts with diffuse channels	RGBA 1-bit alpha			
LoftTunnelBumpSpec.fx	RailWorks specific shader for lofts with diffuse and bump channels generated AO <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Normal		
LoftTunnelTexDiff.fx	RailWorks specific shader for lofts with diffuse channels generated AO	RGBx			
LoftTunnelTexDiffTrans.fx	RailWorks specific shader for lofts with diffuse channels generated AO	RGBA 1-bit alpha			
SkinAmbient.fx	Skinned geometry specific shader Nothing applied, texture mapped looks unfinished	RGBx			
SkinDiffuse.fx	Skinned geometry specific shader with diffuse channel	RGBx			
SkinDiffusePallete2.fx	Skinned geometry specific shader with diffuse channel <u><i>Palettised, see 10.8.3</i></u>	R: palette 1 mask G: palette 2 mask x A: luminosity	RGBA palette 1 256xY image use “_nocomp”	RGBA palette 2 256xY image use “_nocomp”	

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
SkinGloss.fx	Skinned geometry specific shader with diffuse, bump and specular channels <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Normal	Rxxx Specular	
SkinNormal.fx	Skinned geometry specific shader with diffuse and normal channels	RGBx	RGBx Normal		
SkinNormalPallete2.fx	Skinned geometry specific shader with diffuse channel Palettised, see 10.8.3	R: palette 1 mask G: palette 2 mask x A: luminosity	RGBA palette 1 256xY image use “_nocomp”	RGBA palette 2 256xY image use “_nocomp”	RGBx Normal
SkinRESERVED1.fx	Skinned geometry specific shader unfinished	RGBA	xxxx Normal	xxxx Specular	
SkinRESERVED2.fx	Skinned geometry specific shader unfinished	RGBA	xxxx Normal (no forced linear filtering)	xxxx Specular	
SkinRESERVED3.fx	Skinned geometry specific shader unfinished	RGBA	xxxx Normal	xxxx Specular	
SkinSpecular.fx	Skinned geometry specific shader with diffuse channel and specular lighting <i>Supports UVarg1 see 6.3</i>	RGBx			
SpecularBumpmap	unknown	RGBx	RGBx Normal	RGBx Cubemap	
Stars.fx	RailWorks specific shader for sky dome stars	RGBA			
StencilShadow.fx	Stencil shadow objects, material must begin with shadow_ to be detected	xxxx			
SubtractATexDiff	Unknown <i>following naming conventions: Texture mapped, with diffuse colour, using subtractive alpha from texture's alpha channel</i>	RGBA subtractive draw			
Tex	Unlit mapped texture	RGBA			
TexDiff	Texture mapped with single texture, diffuse colour applied	RGBA			
TrainBasicObjectDiffuse.fx	RailWorks specific shader with diffuse channels	RGBx			
TrainBasicObjectUnLit.fx	RailWorks specific shader with diffuse channels. <i>Unlit. Good replacement for Tex.</i>	RGBx			

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
TrainBasicObjectDiffusePallete2.fx	RailWorks specific shader with diffuse channels <i>Palettised, see 10.8.3</i>	R: palette 1 mask G: palette 2 mask x A: luminosity	RGBA palette 1 256xY image use “_nocomp”	RGBA palette 2 256xY image use “_nocomp”	
TrainBasicObjectSpecular.fx	RailWorks specific shader with diffuse channel and specular lighting <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Specular		
TrainBump.fx	RailWorks specific shader with diffuse and bump channels <i>I use it regularly because I don't need Env mapping.</i>	RGBx	RGBx Normal		
TrainBumpEnv.fx	RailWorks specific shader with diffuse and bump channels plus environment map <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Normal	RGBx Cubemap	
TrainBumpEnvMask.fx	RailWorks specific shader with diffuse and bump channels and a static environment mask <i>Supports UVarg1 see 6.3</i>	RGB diffuse A: Envmap	RGBx Normal	RGBx Cubemap	
TrainBumpSpec.fx	<i>RailWorks specific shader with diffuse and bump channels and specular lighting</i> <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Normal		
TrainBumpSpecEnv.fx	RailWorks specific shader with diffuse, specular and bump channels plus environment map <i>Supports UVarg1 see 6.3</i>	RGB diffuse A: Specular	RGBx Normal	RGBx Cubemap	
TrainBumpSpecEnvMask.fx	RailWorks specific shader with diffuse and bump channels and specular lighting and environment map <i>Supports UVarg1 see 6.3</i>	RGB diffuse A: Specular and Envmap	RGBx Normal	RGBx Cubemap	
TrainBumpSpecEnvPalette1.fx	RailWorks specific shader with diffuse, specular and bump channels plus environment map <i>Supports UVarg1 see 6.3</i> <i>Palettised, see 10.8.3</i>	R: palette 1 mask x x A: luminosity and Specular	RGBA palette 1 256xY image use “_nocomp”	RGBx Normal	RGBx Cubemap
TrainBumpSpecMask.fx	RailWorks specific shader with diffuse channel, specular lighting and a static environment mask <i>Supports UVarg1 see 6.3</i>	RGB: diffuse A: Specular	RGBx Normal		
TrainDecal.fx	RailWorks specific shader with special case of decals	RGBA			

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
TrainEmissiveGlow.fx	RailWorks specific shader for glowing emissive materials	RGBA	Rxxx Emissive		
TrainEnv.fx	RailWorks specific shader with diffuse channel and a static environment <i>arg2: Reflection intensity</i>	RGBx	RGBx Cubemap		
TrainEnvMask.fx	RailWorks specific shader with diffuse channel and a static environment mask	RGB: diffuse A: Envmap	RGBx Cubemap		
TrainFlora.fx	RailWorks specific shader with diffuse channels	RGBx			
TrainGlass.fx	Glass shader with env map <i>Supports UVarg1 see 6.3</i> <i>arg2: Reflection intensity</i> <i>arg3: Reflection illumination(night)</i>	RGBA	RGBx Cubemap		
TrainGlassWeatherEffects.fx	Glass shader with weather effects for in cab rain Reflective glass with cubic reflection map and diffuse. <i>See: 8.7.5 - Setting-up TrainGlassWeatherEffects.fx</i>	RGBA	RGBx Cubemap not in BRIAGE	RGBx Object Normal	xxxA Wipers motion
TrainLightBumpSpecMask.fx	RailWorks specific shader with bump map, diffuse and light map channels, specular mask in bump alpha <i>Supports UVarg1 see 6.3</i>	RGBx	RGB: Normal A: Specular	Rxxx LM or AO	
TrainLightMapWithDiffuse.fx	RailWorks specific shader with light map and diffuse channels	RGBx	Rxxx LM or AO		
TrainPostDPPObjectDiffuse.fx	RailWorks specific shader with diffuse channels post DPP stage and shadow mapped	RGBx			
TrainShadowOnly.fx	RailWorks specific empty shader - should never be called, used as a placeholder for shadow only rendering				
TrainShadowOnlyDiffuse.fx	RailWorks specific empty shader - should never be called, used as a placeholder for shadow only rendering with alpha	xxxA 1-bit alpha			
TrainSkyDome.fx	Sky dome shader, dynamically vertex lit by game engine.	xxxx	xxxx		
TrainSpecEnv.fx	RailWorks specific shader with diffuse channel, specular and environment map <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Cubemap		

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
TrainSpecEnvMask.fx	RailWorks specific shader with diffuse channel, specular lighting and a static environment mask <i>Supports UVarg1 see 6.3</i>	RGB: diffuse A: Specular and Envmap	RGBx Cubemap		
TrainUberShader.fx	RailWorks specific shader with diffuse and bump map textures plus a 3rd texture with: specular mask, AO mask and emissive mask <i>Supports UVarg1 see 6.3</i>	RGBx	RGBx Normal	R: LM or AO G: Emissive B: Envmap A: Specular	
TrainUprightViewFacingFlora.fx	RailWorks specific shader for view facing upright flora	RGBx			
TrainVertexLit.fx	Unknown <i>Blends mapped texture with vertex colours. Buggy when skydome behind. No transparency at all.</i>	RGBx			
TrainVertexLitWithDiffuse.fx	Unknown <i>Mapped texture. Buggy when skydome behind.</i>	RGBx			
TrainViewFacingFlora.fx	RailWorks specific shader for view facing flora	RGBx			
TrainViewFacingUnlit.fx	RailWorks specific shader for view facing unlit	RGBx			
TripleGlossMap	Triple texture, 2nd pass contains gloss map in alpha channel, 3rd pass (reflection) texture drawn additively	RGBA	xxxA Gloss	RGBA* Gloss reflection to display additively drawn	
TripleGlossMapWithLightIntens	Triple texture, 2nd pass alpha channel gloss map, 3rd pass drawn additively affected by lightmaps if used	RGBA	xxxA Gloss	RGBA* Gloss reflection to display additively drawn	
TripleGlossMapWithoutLightIntens	Triple texture, 2nd pass alpha channel gloss map, 3rd pass drawn additively not affected by lightmaps if used	RGBA	xxxA Gloss	RGBA* Gloss reflection to display additively drawn	
TripleTexDiffAddAdd	Triple textured, 2nd and 3rd passes are drawn additively	RGBA	RGBA additively drawn	RGBA additively drawn	
TripleTexDiffAlpha	Triple textured, with diffuse colour, using each texture's alpha channels to blend between each pair of passes	RGBA	RGBA	RGBA	
TripleTexDiffAlphaVAlpha	Triple textured, with diffuse colour, pass 2 uses texture alpha for blending, pass 3 uses vertex alpha for blending	RGBA	RGBA	RGBx	

Shader name	Description	Slot 1	Slot 2	Slot 3	Slot 4
TripleTexDiffVAlpha	Triple textured, with diffuse colour, using same vertex alpha to blend between each pair of passes	RGBA	RGBx	RGBx	
TripleTexDiffVAlphaTAlpha	Triple textured, with diffuse colour, pass 2 uses vertex alpha for blending, pass 3 uses texture alpha for blending	RGBA	RGBx	RGBA	
WaterCubeMap.fx	RailWorks specific shader for reflective water <i>Supports UVarg1 see 6.3</i> <i>arg2: 'Wobble' factor</i>	RGBx Normal	xxxx	RGBx Cubemap	
WaterFlow.fx	RailWorks specific shader for reflective water <i>Supports UVarg1 see 6.3</i> <i>arg2: 'Wobble' factor</i>	RGBx Normal	xxxx	RGBx Cubemap	
WaterScenery.fx	RailWorks specific shader for reflective water <i>Supports UVarg1 see 6.3</i> <i>arg2: 'Wobble' factor</i> <i>arg3: Movement speed</i> <i>arg4: Oscillation height</i> <i>arg5: Oscillation scale</i>	RGBx Normal	RGBA	RGBx Cubemap	

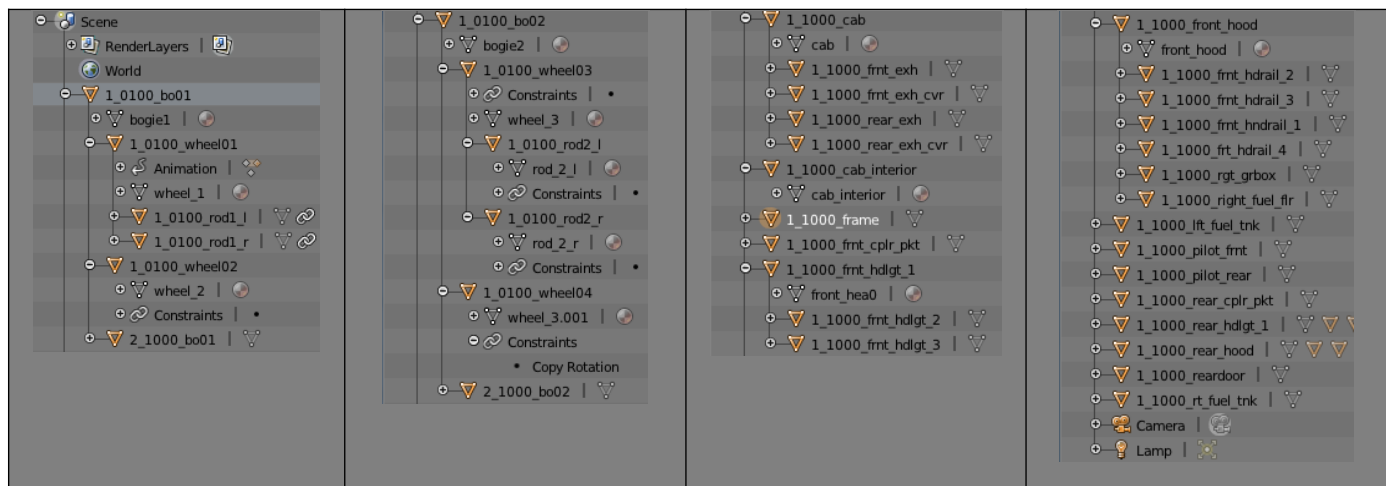
10.2 Shaders usage examples

Use	Shader name	Main texture (slot 1)	Bump map	Cube map	Additional settings
Light halo around lightsources (e.g. streetlights)	AddATex	name.dds (transparency in alpha channel)			→ IGS options <i>ZBufferMode</i> set to <u>Test only</u> → IGS options <i>Viewer facing options</i> set to <u>Viewfacing or UprightViewFacing</u> → <i>Mesh data tab</i> : <u>one vertex colors</u> channel active and renderable. See 8.7.4
Headlight or rearlight glass	AddATex	name.dds (transparency in alpha channel)			→ IGS options <i>ZBufferMode</i> set to <u>Test only</u>
Simple windows only	BlendATexDiff	name.dds (transparency in alpha channel)			→ IGS options <i>ZBufferMode</i> set to <u>Test only</u>
Animated texture <i>see 6.4 - UV Special effects</i>	TexDiff	name_anim1.dds which is the first file of the animation.			→ IGS options <i>AnimateUVs</i> enabled → <i>#Frames</i> and <i>FPS</i> according to the animation to implement.
Scrolling text <i>see 6.4 - UV Special effects</i>	TexDiff or AddATex	name.dds (transparency in alpha channel)			→ IGS options <i>Scroll UVs</i> enabled → <i>u</i> and <i>v</i> according to the speed and direction of scroll to implement. → IGS options <i>ZBufferMode</i> set to <u>Test only</u> if transparency needed
Solid texture with holes	TrainBasicObjectDiffuse.fx	name.dds (transparency in alpha channel. Only black or white (1-bit))			→ IGS options <i>Transparency</i> enabled
Solid texture	TrainBasicObjectDiffuse.fx	name.dds			
Texture with normal map	TrainBump.fx	name.dds	name_nm.dds		
Texture with specular effects and normal maps	TrainBumpSpecEnvMask.fx	name.dds (specular in alpha channel)	name_nm.dds	env.dds	→ <u>UV arguments suggested values</u> : arg1: 32.0 (all other values = 0.0) <i>see 6.3 for more infos</i>

Use	Shader name	Main texture (slot 1)	Bump map	Cube map	Additional settings
Solid texture with holes for inscriptions (such as rolling stock numbers)	TrainDecal.fx	decal_name.dds (transparence in alpha chanel: any grey value between black and white)			→ IGS options <i>ZBufferMode</i> set to <i>Test only</i> (needed for the alpha channel to be properly processed as a transparent layer)
Windows with reflections	TrainGlass.fx	name.dds (transparency in alpha channel)		env.dds	→ <u>UV arguments suggested values</u> : arg1: 64 arg2: 0.8 arg3: 0.4 (all other values = 0.0) <i>see 6.3 for more infos</i>
Texture with specular effects	TrainSpecEnvMask.fx	name.dds (specular in alpha channel)		env.dds	→ <u>UV arguments suggested values</u> : arg1: 32.0 (all other values = 0.0) <i>see 6.3 for more infos</i>
2D Vegetation bushes	TrainUprightViewFacingFlora.fx	name.dds (transparency in alpha channel. 1-bit)			→ IGS options <i>Transparency</i> enabled → <i>Smaller IGS</i> must be checked.(see 4.3) → IGS options <i>Viewer facing options</i> set to <i>Auto</i>
2D Vegetation leaves	TrainViewFacingFlora.fx	name.dds (transparency in alpha channel. 1-bit)			→ IGS options <i>Transparency</i> enabled → <i>Smaller IGS</i> must be checked.(see 4.3) → IGS options <i>Viewer facing options</i> set to <i>Auto</i>

Report to 10.1 - Shaders list for more precise informations.

10.3 Example: Hierarchy and Custom Keywords impact on export



Note: Custom Keywords in this example show how objects are merged and how it can also be used to ensure that an object is not merged with others by adding the full object name (see paragraph also 4.7).

For this example, the wheels don't follow the TS20XX naming rules ([bo01wh01](#), for example). They are processed as any object. With the TS20XX naming rules, it would not have been necessary to add them in the Custom Keywords list (see choices 3 and 4 below).

This model needs to animate wheels and rods. So choice 1 is not the right option as wheels and rods are merged. Choice 4 is the best if you need to add a lot more objects in the model and don't want to reach the 256 objects limit.

IA Export: To export the wheels and rods animation, select **1_0100_bo01** and **1_0100_bo02**.

Note: display of this informations has changed in BRIAGE but may be found at the same location with an increased legibility.

Choice 1: IGS export with:

Hierarchy processing:

enabled

Main object:

1_0100_frame

----- Groups list ----- --- 5 groups --- --- Group --- 1_1000_frame (34 items) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frame 1_1000_cab_interior 1_1000_rear_cplr_pkt 1_1000_frt_cplr_pkt 1_1000_lft_fuel_tnk 1_1000_rt_fuel_tnk 1_1000_pilot_frt 1_1000_pilot_rear 1_1000_cab 1_1000_frt_exh 1_1000_frt_exh_cvr 1_1000_rear_exh 1_1000_rear_exh_cvr 1_1000_rear_hood 1_1000_hndrail_rear_3 1_1000_hndrail_rear_4	1_1000_left_fuel_flr 1_1000_lft_gearbx 1_1000_rear_hndrail_1 1_1000_rear_hndrail_2 1_1000_reardoor 1_1000_rear_hdltg_1 1_1000_rear_hdltg_2 1_1000_rear_hdltg_3 1_1000_front_hood 1_1000_frt_hndrail_2 1_1000_frt_hndrail_3 1_1000_frt_hndrail_1 1_1000_frt_hndrail_4 1_1000_rgt_grbox 1_1000_right_fuel_flr 1_1000_frt_hdltg_1 1_1000_frt_hdltg_2 1_1000_frt_hdltg_3 --- Group --- 1_0100_bo02 (5 items) LOD level 1 / LOD distance 0100 / keyword_bo02 Parent group: None Group objects: 1_0100_bo02	1_0100_wheel03 1_0100_rod2_l 1_0100_rod2_r 1_0100_wheel04 Group children: 2_1000_bo02 --- Group --- 2_1000_bo02 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo02 Parent group: 1_0100_bo02 Group objects: 2_1000_bo02 --- Group --- 1_0100_bo01 (5 items) LOD level 1 / LOD distance 0100 / keyword_bo01 Parent group: None Group objects: 1_0100_bo01 1_0100_wheel01 1_0100_rod1_l 1_0100_rod1_r 1_0100_wheel02	Group children: 2_1000_bo01 --- Group --- 2_1000_bo01 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo01 Parent group: 1_0100_bo01 Group objects: 2_1000_bo01
--	---	--	--

Choice 2: IGS export with:**Hierarchy processing:**

disabled

Main object:

1_0100_frame

----- Groups list ----- --- 46 groups --- --- Group --- 1_1000_frame (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frame --- Group --- 1_0100_bo02 (1 item) LOD level 1 / LOD distance 0100 / keyword bo02 Parent group: None Group objects: 1_0100_bo02 Group children: 1_0100_wheel03 1_0100_wheel04 2_1000_bo02 --- Group --- 1_0100_wheel03 (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_bo02 Group objects: 1_0100_wheel03 Group children: 1_0100_rod2_l 1_0100_rod2_r --- Group --- 1_0100_rod2_l (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_l --- Group --- 1_0100_rod2_r (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_r --- Group --- 1_0100_wheel04 (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_bo02 Group objects: 1_0100_wheel04 --- Group --- 2_1000_bo02 (1 item) LOD level 2 / LOD distance 1000 / keyword bo02 Parent group: 1_0100_bo02 Group objects: 2_1000_bo02 --- Group --- 1_0100_bo01 (1 item) LOD level 1 / LOD distance 0100 / keyword bo01 Parent group: None Group objects: 1_0100_bo01 Group children: 1_0100_wheel01 1_0100_wheel02 2_1000_bo01 --- Group --- 1_0100_wheel01 (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_bo01 Group objects: 1_0100_wheel01 Group children: 1_0100_rod1_l 1_0100_rod1_r --- Group --- 1_0100_rod1_l (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_wheel01 Group objects: 1_0100_rod1_l --- Group --- 1_0100_rod1_r (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_wheel01	Group objects: 1_0100_rod1_r --- Group --- 1_0100_wheel02 (1 item) LOD level 1 / LOD distance 0100 / keyword None Parent group: 1_0100_bo01 Group objects: 1_0100_wheel02 --- Group --- 2_1000_bo01 (1 item) LOD level 2 / LOD distance 1000 / keyword bo01 Parent group: 1_0100_bo01 Group objects: 2_1000_bo01 --- Group --- 1_1000_cab_interior (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_cab_interior --- Group --- 1_1000_rear_cpl_r_pkt (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_cpl_r_pkt --- Group --- 1_1000_frnt_cpl_r_pkt (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frnt_cpl_r_pkt --- Group --- 1_1000_lft_fuel_tnk (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_lft_fuel_tnk --- Group --- 1_1000_rt_fuel_tnk (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rt_fuel_tnk --- Group --- 1_1000_pilot_frnt (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_pilot_frnt --- Group --- 1_1000_pilot_rear (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_pilot_rear --- Group --- 1_1000_cab (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_cab Group children: 1_1000_frnt_exh 1_1000_frnt_exh_cvr 1_1000_rear_exh 1_1000_rear_exh_cvr --- Group --- 1_1000_frnt_exh (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects: 1_1000_frnt_exh --- Group --- 1_1000_frnt_exh_cvr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects:	1_1000_frnt_exh_cvr --- Group --- 1_1000_rear_exh (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects: 1_1000_rear_exh --- Group --- 1_1000_rear_exh_cvr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects: 1_1000_rear_exh_cvr --- Group --- 1_1000_rear_hood (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_hood Group children: 1_1000_hdrail_rear_3 1_1000_hndrail_rear_4 1_1000_left_fuel_flr 1_1000_lft_gearbx 1_1000_rear_hdrail_1 1_1000_rear_hdrail_2 --- Group --- 1_1000_hdrail_rear_3 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_hdrail_rear_3 --- Group --- 1_1000_hndrail_rear_4 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_hndrail_rear_4 --- Group --- 1_1000_left_fuel_flr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_left_fuel_flr --- Group --- 1_1000_lft_gearbx (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_lft_gearbx --- Group --- 1_1000_rear_hdrail_1 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_rear_hdrail_1 --- Group --- 1_1000_rear_hdrail_2 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_rear_hdrail_2 --- Group --- 1_1000_reardoor (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_reardoor --- Group --- 1_1000_rear_hdlgt_1 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_hdlgt_1 Group children: 1_1000_rear_hdlgt_2 1_1000_rear_hdlgt_3 --- Group --- 1_1000_rear_hdlgt_2 (1 item) LOD level 1 / LOD distance 1000 / keyword None	Parent group: 1_1000_rear_hdlgt_1 Group objects: 1_1000_rear_hdlgt_2 --- Group --- 1_1000_rear_hdlgt_3 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hdlgt_1 Group objects: 1_1000_rear_hdlgt_3 --- Group --- 1_1000_front_hood (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_front_hood Group children: 1_1000_frnt_hdrail_2 1_1000_frnt_hdrail_3 1_1000_frnt_hdrail_1 1_1000_frnt_hdrail_4 1_1000_rgt_grbox 1_1000_right_fuel_flr --- Group --- 1_1000_frnt_hdrail_2 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_frnt_hdrail_2 --- Group --- 1_1000_frnt_hdrail_3 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_frnt_hdrail_3 --- Group --- 1_1000_frnt_hdrail_1 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_frnt_hdrail_1 --- Group --- 1_1000_frnt_hdrail_4 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_frnt_hdrail_4 --- Group --- 1_1000_rgt_grbox (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_rgt_grbox --- Group --- 1_1000_right_fuel_flr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_right_fuel_flr --- Group --- 1_1000_frnt_hdlgt_1 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frnt_hdlgt_1 Group children: 1_1000_frnt_hdlgt_2 1_1000_frnt_hdlgt_3 --- Group --- 1_1000_frnt_hdlgt_2 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_frnt_hdlgt_1 Group objects: 1_1000_frnt_hdlgt_2 --- Group --- 1_1000_frnt_hdlgt_3 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_frnt_hdlgt_1 Group objects: 1_1000_frnt_hdlgt_3
---	---	--	--

Choice 3: IGS export with:

Hierarchy processing:

disabled

Main object:

1_0100_frame

Custom Keywords:

frnt	1_0100_rod2_l	1_0100_wheel03
rail	1_0100_rod2_r	1_0100_wheel04
1_0100_rod1_l	1_0100_wheel01	
1_0100_rod1_r	1_0100_wheel02	

----- Groups list ----- --- 33 groups --- --- Group --- 1_1000_frame (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frame --- Group --- 1_0100_bo02 (1 item) LOD level 1 / LOD distance 0100 / keyword_bo02 Parent group: None Group objects: 1_0100_bo02 Group children: 1_0100_wheel03 1_0100_wheel04 2_1000_bo02 --- Group --- 1_0100_wheel03 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel03 Parent group: 1_0100_bo02 Group objects: 1_0100_wheel03 Group children: 1_0100_rod2_l 1_0100_rod2_r --- Group --- 1_0100_rod2_l (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod2_l Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_l --- Group --- 1_0100_rod2_r (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod2_r Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_r --- Group --- 1_0100_wheel04 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel04 Parent group: 1_0100_bo02 Group objects: 1_0100_wheel04 --- Group --- 2_1000_bo02 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo02 Parent group: 1_0100_bo02 Group objects: 2_1000_bo02 --- Group --- 1_0100_bo01 (1 item) LOD level 1 / LOD distance 0100 / keyword_bo01 Parent group: None Group objects: 1_0100_bo01 Group children: 1_0100_wheel01 1_0100_wheel02	2_1000_bo01 --- Group --- 1_0100_wheel01 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel01 Parent group: 1_0100_bo01 Group objects: 1_0100_wheel01 Group children: 1_0100_rod1_l 1_0100_rod1_r --- Group --- 1_0100_rod1_l (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod1_l Parent group: 1_0100_wheel01 Group objects: 1_0100_rod1_l --- Group --- 1_0100_rod1_r (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod1_r Parent group: 1_0100_wheel01 Group objects: 1_0100_rod1_r --- Group --- 1_0100_wheel02 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel02 Parent group: 1_0100_bo01 Group objects: 1_0100_wheel02 --- Group --- 2_1000_bo01 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo01 Parent group: 1_0100_bo01 Group objects: 2_1000_bo01 --- Group --- 1_1000_cab_interior (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_cab_interior --- Group --- 1_1000_rear_cpl_r_pkt (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_cpl_r_pkt --- Group --- 1_1000_frnt_cpl_r_pkt (10 items) LOD level 1 / LOD distance 1000 / keyword frnt Parent group: None Group objects: 1_1000_frnt_cpl_r_pkt 1_1000_pilot_frnt 1_1000_frnt_exh 1_1000_frnt_exh_cvr 1_1000_frnt_hdrail_2 1_1000_frnt_hdrail_3 1_1000_frnt_hdrail_1 1_1000_frnt_hdlgt_1 1_1000_frnt_hdlgt_2	1_1000_frnt_hdlgt_3 --- Group --- 1_1000_lft_fuel_tnk (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_lft_fuel_tnk --- Group --- 1_1000_rt_fuel_tnk (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rt_fuel_tnk --- Group --- 1_1000_pilot_rear (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_pilot_rear --- Group --- 1_1000_cab (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_cab Group children: 1_1000_rear_exh 1_1000_rear_exh_cvr --- Group --- 1_1000_rear_exh (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects: 1_1000_rear_exh --- Group --- 1_1000_rear_exh_cvr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_cab Group objects: 1_1000_rear_exh_cvr --- Group --- 1_1000_rear_hood (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_hood Group children: 1_1000_hdrail_rear_3 1_1000_left_fuel_flr 1_1000_lft_gearbx --- Group --- 1_1000_hdrail_rear_3 (5 items) LOD level 1 / LOD distance 1000 / keyword rail Parent group: 1_1000_rear_hood Group objects: 1_1000_hdrail_rear_3 1_1000_hdrail_rear_4 1_1000_rear_hdrail_1 1_1000_rear_hdrail_2 1_1000_frnt_hdrail_4	--- Group --- 1_1000_left_fuel_flr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_left_fuel_flr --- Group --- 1_1000_lft_gearbx (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hood Group objects: 1_1000_lft_gearbx --- Group --- 1_1000_reardoor (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_reardoor --- Group --- 1_1000_rear_hdlgt_1 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_rear_hdlgt_1 Group children: 1_1000_rear_hdlgt_2 1_1000_rear_hdlgt_3 --- Group --- 1_1000_rear_hdlgt_2 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hdlgt_1 Group objects: 1_1000_rear_hdlgt_2 --- Group --- 1_1000_rear_hdlgt_3 (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_rear_hdlgt_1 Group objects: 1_1000_rear_hdlgt_3 --- Group --- 1_1000_front_hood (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_front_hood Group children: 1_1000_rgt_grbox 1_1000_right_fuel_flr --- Group --- 1_1000_rgt_grbox (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_rgt_grbox --- Group --- 1_1000_right_fuel_flr (1 item) LOD level 1 / LOD distance 1000 / keyword None Parent group: 1_1000_front_hood Group objects: 1_1000_right_fuel_flr
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Choice 4: IGS export with:

Hierarchy processing:

enabled

Main object:

1_0100_frame

Custom Keywords:

frnt	1_0100_rod2_l	1_0100_wheel03
rail	1_0100_rod2_r	1_0100_wheel04
1_0100_rod1_l	1_0100_wheel01	
1_0100_rod1_r	1_0100_wheel02	

----- Groups list ----- --- 15 groups --- --- Group --- 1_1000_frame (19 items) LOD level 1 / LOD distance 1000 / keyword None Parent group: None Group objects: 1_1000_frame 1_1000_cab_interior 1_1000_rear_cpl_r_pkt 1_1000_lft_fuel_tnk 1_1000_rt_fuel_tnk 1_1000_pilot_rear 1_1000_cab 1_1000_rear_exh 1_1000_rear_exh_cvr 1_1000_rear_hood 1_1000_left_fuel_flr 1_1000_lft_gearbx 1_1000_reardoor 1_1000_rear_hdlgt_1 1_1000_rear_hdlgt_2 1_1000_rear_hdlgt_3 1_1000_front_hood 1_1000_rgt_grbox 1_1000_right_fuel_flr	--- Group --- 1_0100_wheel03 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel03 Parent group: 1_0100_bo02 Group objects: 1_0100_wheel03 Group children: 1_0100_rod2_l 1_0100_rod2_r	LOD level 1 / LOD distance 0100 / keyword_bo01 Parent group: None Group objects: 1_0100_bo01 Group children: 1_0100_wheel01 1_0100_wheel02 2_1000_bo01	1_0100_wheel02 --- Group --- 2_1000_bo01 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo01 Parent group: 1_0100_bo01 Group objects: 2_1000_bo01
	--- Group --- 1_0100_rod2_l (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod2_l Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_l	--- Group --- 1_0100_wheel01 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel01 Parent group: 1_0100_bo01 Group objects: 1_0100_wheel01 Group children: 1_0100_rod1_l 1_0100_rod1_r	--- Group --- 1_1000_frnt_cpl_r_pkt (10 items) LOD level 1 / LOD distance 1000 / keyword frnt Parent group: None Group objects: 1_1000_frnt_cpl_r_pkt 1_1000_pilot_frnt 1_1000_frnt_exh 1_1000_frnt_exh_cvr 1_1000_frnt_hdrail_2 1_1000_frnt_hdrail_3 1_1000_frnt_hdrail_1 1_1000_frnt_hdlgt_1 1_1000_frnt_hdlgt_2 1_1000_frnt_hdlgt_3
	--- Group --- 1_0100_rod2_r (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod2_r Parent group: 1_0100_wheel03 Group objects: 1_0100_rod2_r	--- Group --- 1_0100_rod1_l (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod1_l Parent group: 1_0100_wheel01 Group objects: 1_0100_rod1_l	--- Group --- 1_1000_hdrail_rear_3 (5 items) LOD level 1 / LOD distance 1000 / keyword rail Parent group: None Group objects: 1_1000_hdrail_rear_3 1_1000_hdrail_rear_4 1_1000_rear_hdrail_1 1_1000_rear_hdrail_2 1_1000_frnt_hdrail_4
	--- Group --- 1_0100_wheel04 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel04 Parent group: 1_0100_bo02 Group objects: 1_0100_wheel04	--- Group --- 1_0100_rod1_r (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_rod1_r Parent group: 1_0100_wheel01 Group objects: 1_0100_rod1_r	
	--- Group --- 2_1000_bo02 (1 item) LOD level 2 / LOD distance 1000 / keyword_bo02 Parent group: 1_0100_bo02 Group objects: 2_1000_bo02	--- Group --- 1_0100_wheel02 (1 item) LOD level 1 / LOD distance 0100 / keyword 1_0100_wheel02 Parent group: 1_0100_bo01 Group objects:	
	--- Group --- 1_0100_bo01 (1 item)		

10.4 TS20XX naming rule check

All objects should follow strict naming conventions.

Each name starts with a single digit representing the LOD level, followed by a 4-digit visible distance between underscores. After this a logically chosen object name follows and the whole name is limited to a maximum of 31 characters.

From Railworkswiki

Operations are performed after all name modifications have been applied.

The exporter checks the beginning of the name looking for a digit (LOD level), an underscore, 4 digits (visible distance), an underscore and the name.

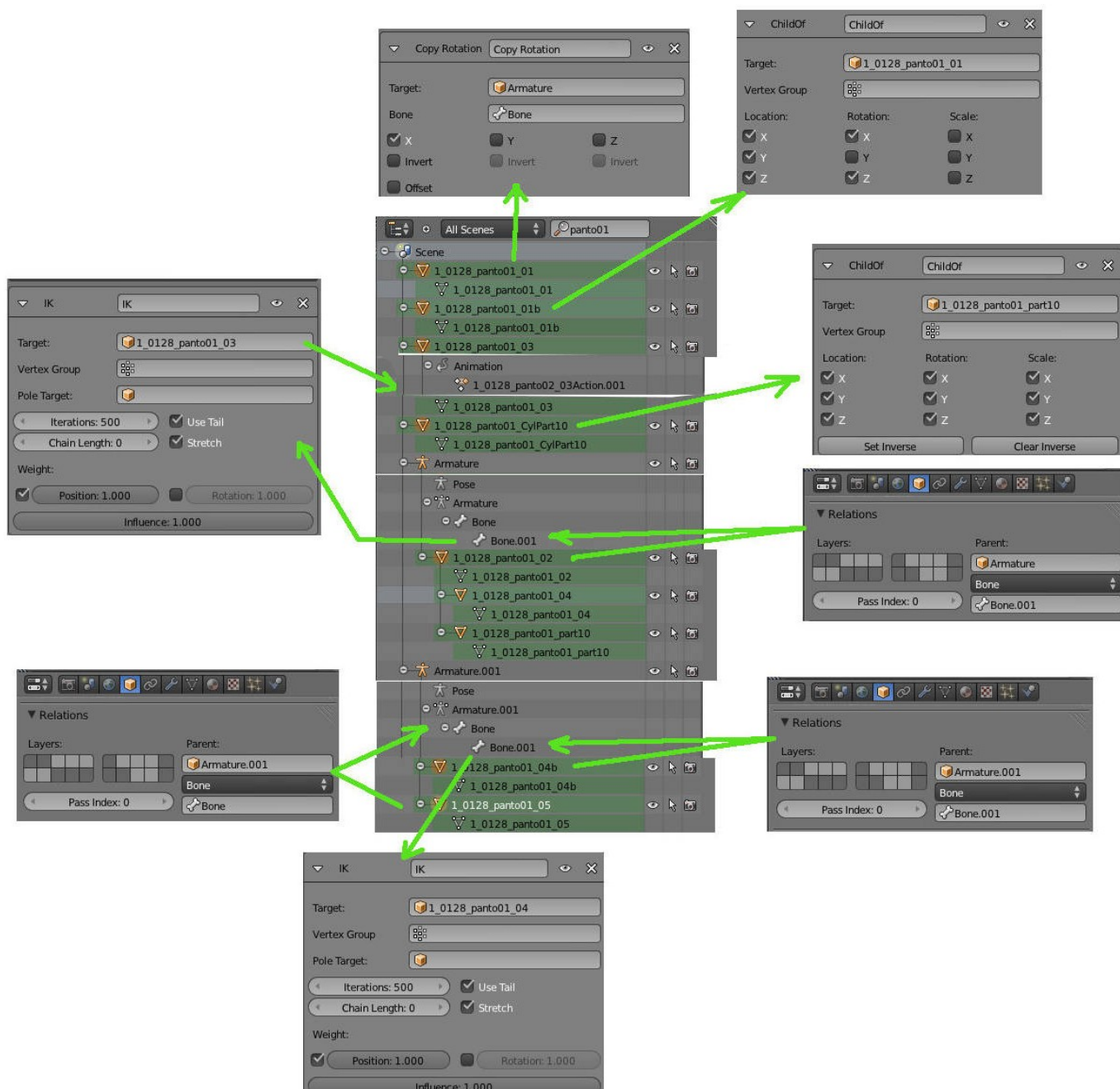
If the exporter doesn't find 1 digit, then an underscore, then a number, then an underscore, the name will be automatically converted to LOD = 1, distance = 1000, followed by the original name. For example, a Blender object named "MyObject" or "1-0100_MyObject" will be converted as, respectively, "1_1000_MyObject" or "1_1000_1-0100_MyObject".

Otherwise, a best fit to the number is found. For example:

- 1_02_object is converted as 1_0200_object
- 1_2_object is converted as 1_2000_object
- 2_10000_object is converted as 2_1000_object

Warning messages are written to the log file about misnamed objects which name is converted.

10.5 Selection example for IA export



These should be selected at least:

Object	Raison for selection
1_0128_panto01_03	Object without parent (no parentage created with Ctrl p). It has an animation.
1_0128_panto01_02	Object which parent is an armature
1_0128_panto01_CylPart10	Object without parent (no parentage created with Ctrl p)
1_0128_panto01_01	Object without parent (no parentage created with Ctrl p)
1_0128_panto01_01b	Object without parent (no parentage created with Ctrl p)
1_0128_panto01_04b	Object which parent is an armature
1_0128_panto01_05	Object which parent is an armature

The other *panto01* objects like 1_0128_panto01_04 don't need to be selected because they are children of already selected objects.

Other objects not shown here (little additional parts) don't need to be selected either because they are children of *panto01*. They will be present in the exported animation.

10.6 Dynamic text

Please refer to the excellent tutorial on:

https://www.christrains.com/en/ts_faq_autonumbering.html (The paragraph about "Multi/Sub-Object material" is specific to 3DSMax.)

Please note that it may be necessary to import again the equipment in a scenario so that the game takes into account changes in xml files or in the Blender model.

In addition to this tutorial, here is how 3 numbering objects with 3 digits were implemented in Blender:

They are respectively named 1_0050_primarydigits_3 (for the number on the right of the image), 1_0050_primarydigits_3.001 (for the number on the opposite side) and 1_0050_primarydigits_3.002 (for the front number).

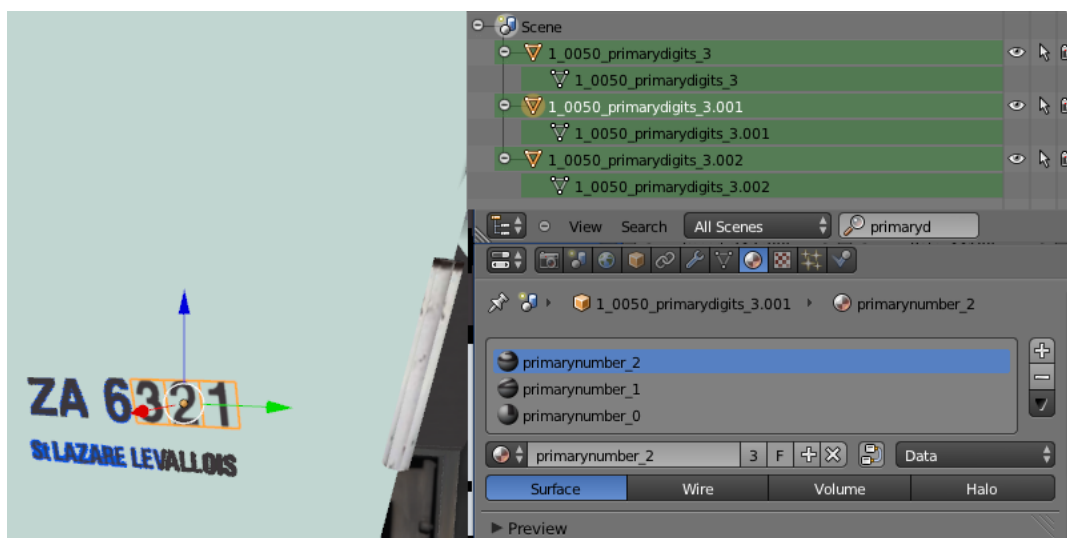


The white rectangle shows the main bicolour texture and the alpha texture for digit 3 included in a 64 x 128 pixels DDS file. The faces for the side grey number crop the grey area and the faces for the front white number crops the white area.

The text is set here to "467".

The 1_0050_primarydigits_3 object (1_yyyy_primarydigits_ followed by the number of digits = 3) is a plane divided into 3 faces. The other 1_0050_primarydigits_3.00x objects are created by duplication.

Each face is assigned a material: primarynumber_0 (units), primarynumber_1 (tens digit) ou primarynumber_2 (hundreds digit).



In detail, each of these 3 materials is defined by (N is set to 0, 1, 2, ..., 29, ..., 99, and perhaps even higher):

- **Material name:** primarynumber_N
- **Image in UV-editor :** decal_primarynumber_N.dds
- **Selected shader:** TrainDecal.fx
- **Material first texture slot:** texture named decal_primarynumber_N associated in the UV-editor to image decal_primarynumber_N.dds and associated to texture file decal_primarynumber_N.dds.

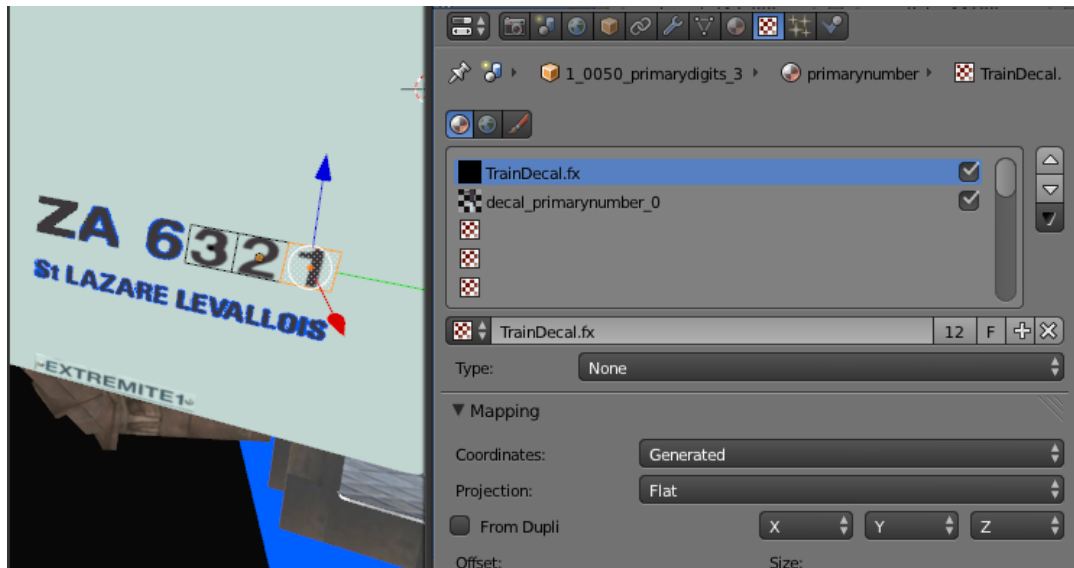
File decal_primarynumber_0.dds is created by copying and renaming *number_1.dds* (thus digit 1 is displayed in the above screenshot. This digit will be replaced in game by the corresponding value read in the csv file).

File *decal_primarynumber_1.dds* is created by copying and renaming *number_2.dds* (thus digit 2 is displayed in the above screenshot).

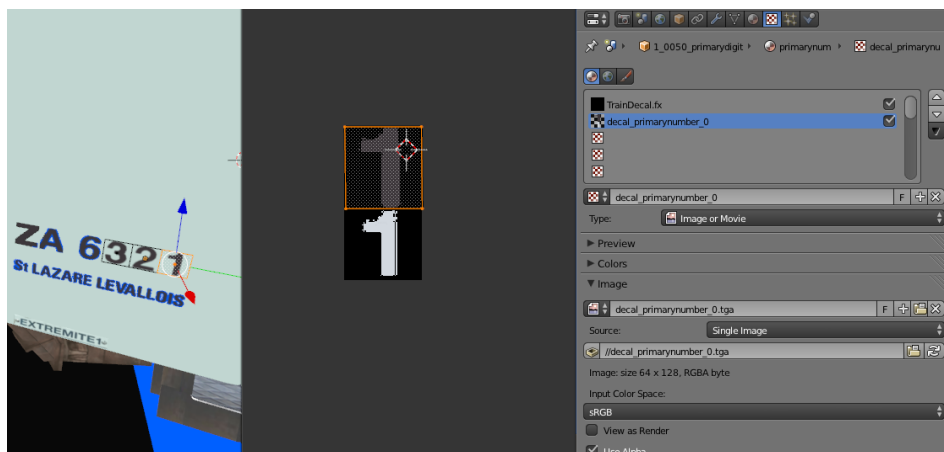
File *decal_primarynumber_2.dds* is created by copying and renaming *number_3.dds* (thus digit 3 is displayed in the above screenshot).

For example for *primarynumber_0*:

Material first texture slot:



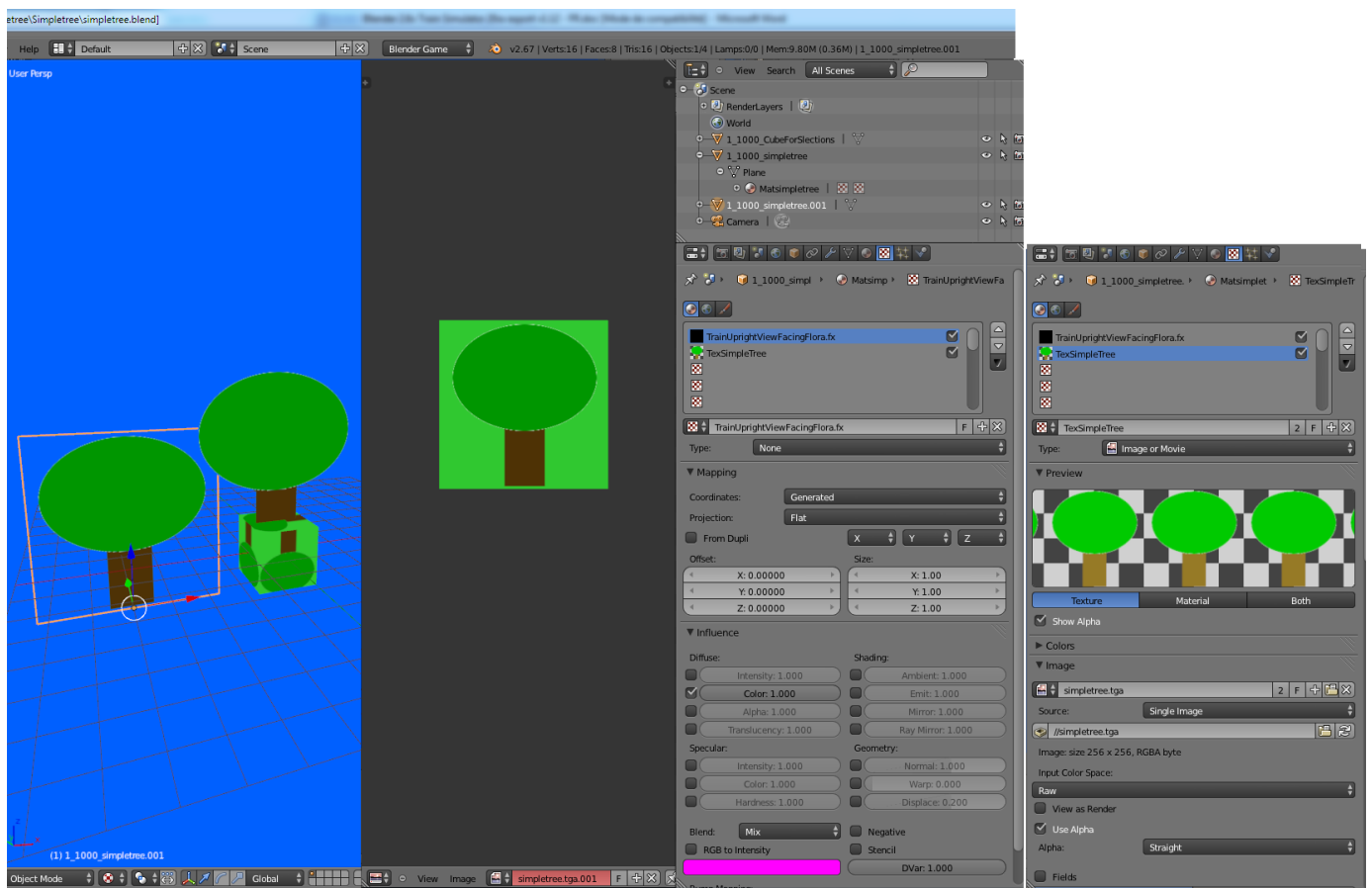
Material second texture slot (in the UV-editor, the image is *decal_primarynumber_0.dds*)



Nota: Be careful when creating `NamedTextureSets` for your dynamic text. `TextureName` field of each entry must contain *1 ASCII symbol*, else your text won't display at all.

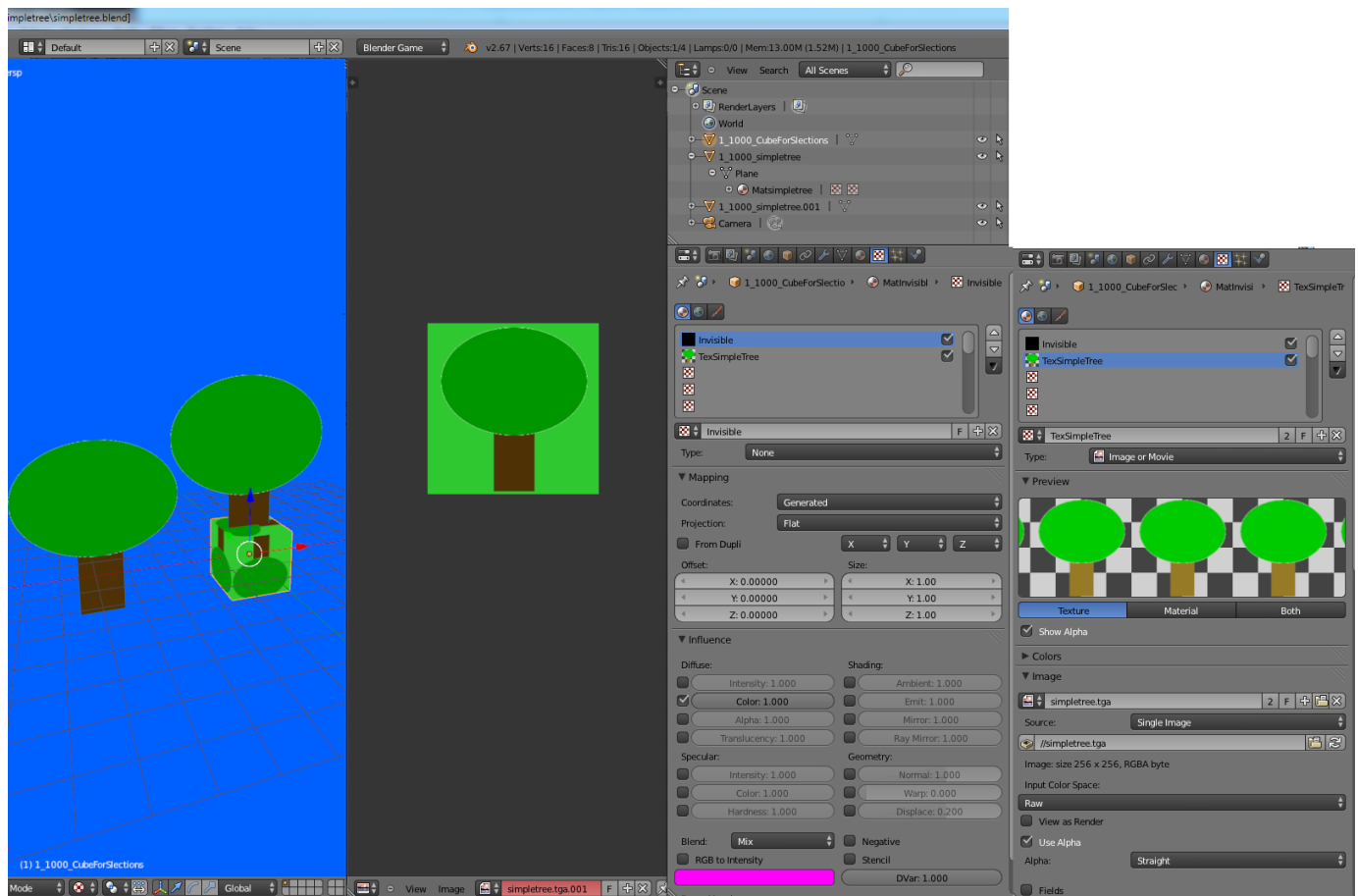
10.7 Example with *TrainUprightViewFacingFlora.fx*

Tree texture in Material "MatSimpletree" with "Transparency" enabled in IGS option panel:



(Texture file provided by cilldroichid)

Texture for the selection cube (Material "MatInvisible"). An invisible object must be added otherwise it will be impossible to select the object once placed in a scenario. There is no special care to take to texture this item because it will not be visible.



Result:



10.8 Texturing lexical notes

10.8.1 3 components maps (RGB)

Diffuse map (*Diff* or *albedo*) contains **colour** informations.

Normal map (or just *Normal*) holds informations about fake **details**. i.e. , details too heavy and too little to be included in final 3D work are printed to a normal map to spare performances. This can be useful for vents, screws, metal panels, hinges, windows, etc...

Cubic map (*Cubemap* or *Skybox*) is a special texture type used to **fake reflections** on old rendering engines. It contains 6 images mapped inside a virtual cube. They should look like the real mirrored environment around your object for more realism. These can be contained in DDS files; use GIMP or Ps to be able to manipulate them.

10.8.2 1 component maps (greyscale)

Transparency is holding **opacity** informations, it is common to find it nicknamed as Alpha because it is often stored in Alpha channel of a diffuse texture. Its value blends current texture with what stands behind.

Specular map (or *Spec*) specifies **specular intensity** (“specular power” in TS: see 6.2) at given places of the texture . This is used to fake a particular real life material light properties, making it shiny or dull.

Gloss map (or simply *Gloss*) specifies **specular size** (“Phong” size in TS: see 6.3, *Hardness* in Blender, “tightness”) at given places of the texture . This is used to fake a particular real life material light properties, making it glossy or diffuse.

Environment map (or *Envmap*) holds **reflectiveness** of a material making it more or less mirror-like.

Light map (*LM*), which save shadows from all light sources, and Ambient occlusion (*AO*) map, which save shadows from bouncing environmental light (in more or less fake way), can be considered the same in TS as they are treated the same way. Both contain fixed **shadows** which are altered by dynamic TS light sources and occlusion volumes. However, you can mix them to your diffuse maps or use them as stencil shadows (e.g. with SubtractATexDiff) to make them unalterable by game engine.

Emissive map specifies which part of a material **emits light** and in which amount. This can be used to make glowing effects or to cut off a light build into a wall texture.

Greyscale maps sometimes need to be cast to RGB components (just copy values to all channels) when TS needs it.

10.8.3 Palettisation

This refers to a semi-random way of generating variations for diffuse channels in TS.
Take a look at this document:

'*TS directory*'/dev/Docs/TS2014Tech_P1.pdf, p19, 6. *Palettisation*

You may also see and try using the provided **PalletisedCar.blend** file (Thanks to Ash from UKTS) and it's associated blueprints



Maintenance and contact infos

Last modification:

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Made with love, dedication and boredom by your servant Juju49.

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